

# CellScientist: From Execution Feedback to Auditable Model-Revision Trajectories for Cellular Perturbation Prediction

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## ABSTRACT

Cellular perturbation-response modeling requires coordinated choices of representations, fusion mechanisms, objectives, and training procedures. Large language models (LLMs) can propose executable candidates, but unconstrained revision can produce invalid implementations, change task semantics, or discard useful components. We present **CellScientist**, a protocol-constrained workflow that converts execution and validation feedback into auditable model-revision trajectories. It records design states and outcomes, routes discrepancies to specific components, and applies local revisions under a fixed task contract. Cell Painting benchmarks establish the predictive utility of its task-specific, validation-selected models. A separate controlled audit fixes the candidate language, predictor, fitting procedure, and evaluation budget to isolate controller behavior. These comparisons support the value of feedback-guided revision, while registered operational audits examine history reuse, contract checks, and discrepancy routing. Transcriptomic and single-cell experiments extend the evaluation to additional response spaces. Detailed trajectories retain improvements, regressions, rejected candidates, and execution failures, making the evidence behind model selection inspectable. Project page: <https://limengran98.github.io/CellScientist/>.

## 1 INTRODUCTION

Cellular perturbation-response assays measure how chemical or genetic interventions reshape cellular states. Complementary readouts include Cell Painting morphology, transcriptomic profiles, and single-cell measurements (Stirling et al., 2021; Seal et al., 2025; Subramanian et al., 2017). Predicting these responses can extend observed effects to unmeasured perturbations and contexts, supporting compound characterization, mechanism-of-action analysis, and virtual-cell modeling.

Generalization remains difficult because measured responses combine perturbation-induced biology with assay- and context-specific variation. Morphological profiles depend on plate and processing factors (Arevalo et al., 2024), while transcriptomic responses vary across batches and cellular contexts (Zhou et al., 2025; Zhang et al., 2025). Effective predictors therefore require coordinated choices of perturbation and context representations, dose conditioning, feature interactions, objectives, and training procedures. Revising one component can change the behavior of others, making feedback attribution central to model development.

These dependencies create a feedback-attribution problem. A poor validation outcome does not identify whether the next change should target conditioning, representation, fusion, or the objective. Execution failures raise a different question: whether a candidate can be evaluated under the intended protocol. A useful search controller must distinguish these signals before converting them into a concrete revision.

AutoML systems search predefined model and hyperparameter spaces (Wang et al., 2021; Gijbbers et al., 2024), whereas LLM-based modeling agents can generate and revise executable programs (Jiang et al., 2025; Lu et al., 2024; Tang et al., 2025). This flexibility also requires control over how execution and

validation outcomes guide subsequent changes. Revisions must preserve task semantics, target components implicated by the feedback, and retain the evidence behind each decision. We therefore ask how execution feedback can guide localized model-design revisions while preserving a fixed task contract.

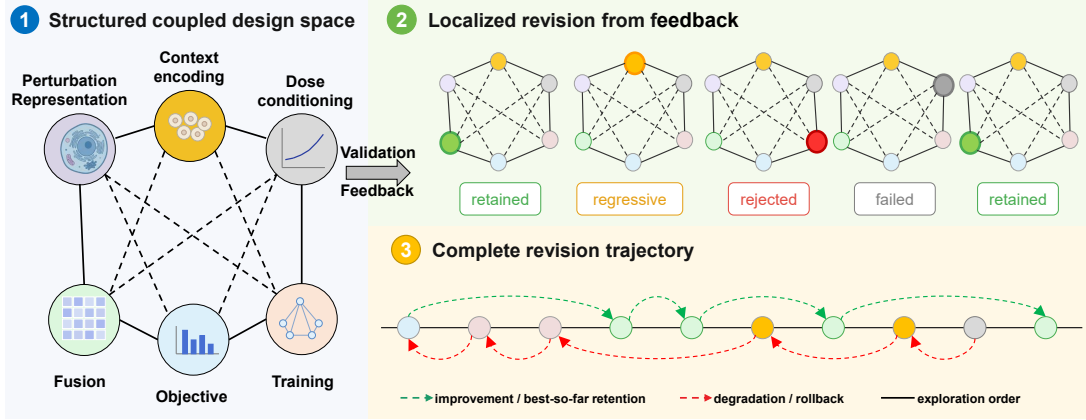


Figure 1: **Conceptual overview of CellScientist.** Revision operates over coupled choices of perturbation representation, context encoding, dose conditioning, fusion, objectives, and training strategy. Recorded feedback guides local updates, while the trajectory preserves improvements, regressions, rollbacks, rejected candidates, execution failures, and validation-retained states.

We propose **CellScientist**, a framework for structured and auditable model revision in cellular perturbation-response prediction. It represents the coupled design space, routes feedback to implicated components, and records successive revisions (Figure 1). Each state links a design, implementation, and outcome under a fixed task contract that preserves data, interface, and evaluation semantics.

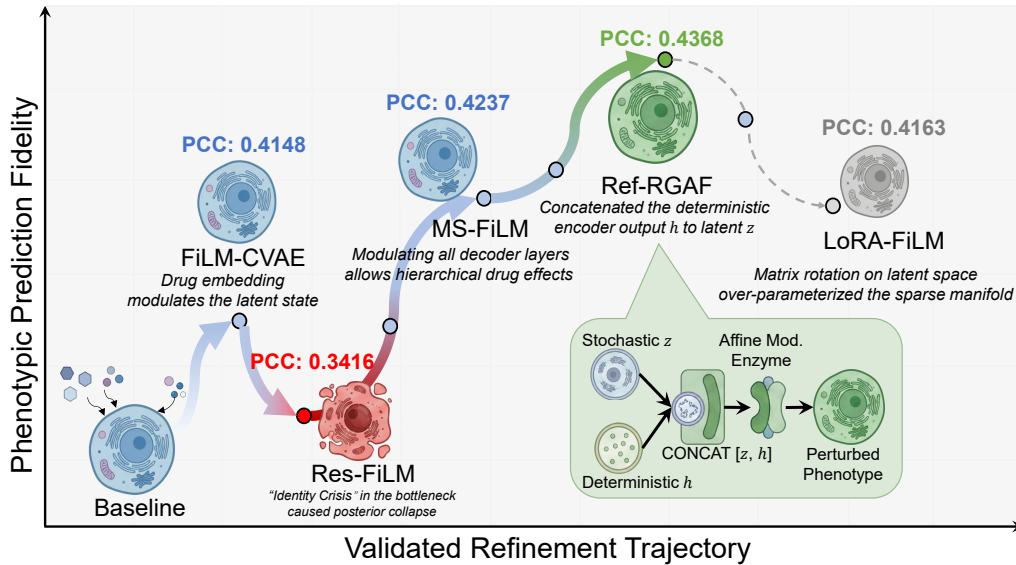


Figure 2: **Feedback-guided model-revision trajectory on BBBC047.** Recorded validation feedback guides successive revisions from modulation-based candidates to Ref-RGAF, which adds a deterministic control-reference encoding and attains the highest validation PCC in this trajectory. The annotations summarize working diagnostics recorded during the search. The retained record includes intermediate regressions and the later, lower-performing LoRA-FiLM candidate.

Figure 2 illustrates this process on BBBC047. Modulation-based candidates show both improvements and regressions, prompting a working diagnosis that the supplied control-reference information is under-used. A localized revision introduces a deterministic reference encoding alongside the stochastic latent

state, yielding the validation-selected Ref-RGAF candidate. The recorded later regression makes clear why the selected model differs from the last attempted candidate. Appendix B.2 gives the detailed trace.

CellScientist makes this transition explicit through three complementary functions. History-Aware Revision Tracking (HRT) maintains design states and evidence from previous attempts. Limited-Change Application (LCA) checks candidate eligibility and applies bounded repair. Performance-Discrepancy Refinement (PDR) links observed discrepancies to revision targets. Together, they connect the rationale for an edit to its implementation and subsequent validation, while preserving unsuccessful branches for inspection.

Our contributions are threefold. First, we formulate a *model-design revision-routing problem* over coupled components under a fixed task contract. Second, CellScientist links local revisions to auditable design states, implementations, and outcomes. Third, task-specific searches evaluate predictive utility, while matched-budget comparisons and component audits isolate controller behavior under held-out reporting.

## 2 RELATED WORK

**Cellular perturbation-response modeling.** Perturbation-response modeling spans morphology, transcriptomic, and single-cell readouts. Cell Painting models must represent chemical structure, dose, experimental context, and correlated morphological targets (Stirling et al., 2021; Chandrasekaran et al., 2023; Seal et al., 2025). DeepCE, MultiDCP, and TranSiGen address transcriptomic response prediction (Pham et al., 2021; Wu et al., 2022; Tong et al., 2024); single-cell benchmarks cover combinatorial gene activation and joint RNA–protein responses (Norman et al., 2019; Papalexi et al., 2021). MVCBench evaluates drug and gene representations across transcriptomic and morphological phenotypes (Li et al., 2026a). CHMR and MORSE integrate molecular structure with cellular and genomic modalities through hierarchical or structured semantic modeling (Li et al., 2026b; Zhang et al., 2026). CellScientist addresses how task-specific predictors are revised and selected under a fixed evaluation contract.

**Automated model development.** Automated model development ranges from search within predefined spaces to revision of generated programs. FLAML and random-search benchmarks automate model and hyperparameter selection (Wang et al., 2021; Gijssbers et al., 2024); LLM-based agents extend the proposal space through code generation and iterative revision (Jiang et al., 2025; Lu et al., 2024; Turcan et al., 2025). Because these approaches differ in model families and compute requirements, predictive comparisons alone cannot isolate controller quality. We therefore separate heterogeneous task-specific benchmarks from a controlled audit with a shared candidate language and matched evaluation budgets.

**Biomedical agents and virtual cells.** Biomedical agents support spatial biology (Wang et al., 2025a; Yu et al., 2025), omics analysis (Qiu et al., 2025), and protocol automation (Wang et al., 2025b). Their performance also depends on workflow structure and reproducible evaluation (Miller et al., 2025). CellForge studies agent-designed perturbation models (Tang et al., 2025), and VCWorld integrates structured biological knowledge into virtual-cell simulation (Wei et al., 2026). HarmonyCell combines hierarchical, validation-guided search with execution pruning, repair, and solution reuse for single-cell perturbation modeling (Huang et al., 2026). CellScientist focuses on refinement control: maintaining explicit design states, applying local revisions under protected task semantics, and routing execution or validation discrepancies to relevant components.

## 3 METHODOLOGY

**CellScientist** is a protocol-constrained controller for iterative perturbation-response model search. It links a model-design space  $\mathcal{H}$  to an executable-candidate space  $\mathcal{I}$  through the loop “Design Specification  $\rightarrow$  Executable Candidate  $\rightarrow$  Validation Feedback  $\rightarrow$  Local Revision,” with  $\mathbf{h}$  denoting a model-design state.

### 3.1 TASK CONTRACT AND ITERATIVE SEARCH

Let  $\mathbf{h}_t \in \mathcal{H}$  denote the model-design specification at iteration  $t$  and let  $\mathbf{c}_t \in \mathcal{I}$  denote an executable candidate. A task is specified by  $\mathcal{D}_{\text{spec}} = (\mathcal{X}, \mathcal{P}, \mathcal{Y}, \mathcal{S}, \mathcal{E})$ , where  $\mathcal{X}$  contains observed cellular context,

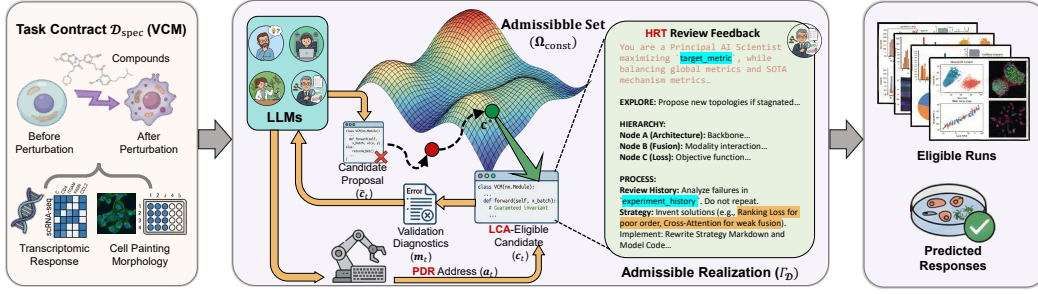


Figure 3: **Overview of the CellScientist framework.** The task contract  $\mathcal{D}_{\text{spec}}$  fixes inputs, perturbation descriptors, response targets, data partitions, and evaluation semantics. HRT supplies the design state  $\mathbf{h}_t$ , task-state graph  $\mathcal{G}_t$ , and run history  $\mathcal{B}_t$  for proposing  $\tilde{\mathbf{c}}_t$ . LCA applies configured checks and bounded repair to return an eligible candidate  $\mathbf{c}_t$  or rejection  $\perp$ . Validation yields diagnostics  $\mathbf{m}_t$ , which PDR maps to a component address  $\mathbf{a}_t$  for localized revision. Validation selection retains the best eligible candidate; the complete trajectory is recorded.

$\mathcal{P}$  contains perturbation descriptors,  $\mathcal{Y}$  contains response targets,  $\mathcal{S}$  defines the data-partition protocol, and  $\mathcal{E}$  defines the evaluation functional. The task contract remains fixed throughout a search trajectory.

The controller implements the following operational sequence:

$$\begin{aligned} \tilde{\mathbf{c}}_t &= \Phi(\mathbf{h}_t, \mathcal{G}_t, \mathcal{B}_t), \\ \mathbf{c}_t &= \Gamma_{\mathcal{D}}(\tilde{\mathbf{c}}_t; \mathbf{h}_t, \mathcal{G}_t, \mathcal{B}_t, R) \in \Omega_{\text{const}} \cup \{\perp\}, \\ \mathbf{m}_t &= \mathcal{F}(\mathbf{c}_t; \mathcal{D}_{\text{val}}, \mathcal{E}), \\ \mathbf{h}_{t+1} &= \mathbf{h}_t \oplus \mathcal{T}_{\text{edit}}(\mathcal{A}(\mathbf{m}_t, \mathcal{G}_t, \mathcal{B}_t) \mid \mathbf{m}_t). \end{aligned} \quad (1)$$

Here  $\Phi$  is the LLM-conditioned proposal operator,  $\Gamma_{\mathcal{D}}$  is the configured check/repair/rejection procedure,  $\mathbf{m}_t$  records execution and validation diagnostics, and  $\mathcal{A}$  selects the intended revision target. The controller state is  $\mathcal{Z}_t = (\mathbf{h}_t, \mathcal{G}_t, \mathcal{B}_t)$ , where  $\mathcal{G}_t$  is the structured task-state record and  $\mathcal{B}_t$  is the run history containing attempted implementations and outcomes. Equation 1 summarizes the implemented sequential search procedure.

### 3.2 HRT: HISTORY-AWARE REVISION TRACKING

History-Aware Revision Tracking (HRT) represents the editable model-design state as a directed task-state graph  $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t)$ . Nodes denote components such as perturbation representation, cellular-context encoding, fusion, architecture, objective design, and training strategy; edges record configured dependencies or ordering relations. The graph provides structured context for candidate generation and is updated with task status and evidence from prior attempts.

Given a diagnostic, the controller records one or more intended revision targets in  $\mathcal{V}_t$  and supplies their component descriptions to the next proposal step, making the edit scope and history inspectable. HRT also carries configured task rules, including required stages, interface expectations, and protected evaluation fields, for downstream checks.

### 3.3 LCA: LIMITED-CHANGE APPLICATION

Limited-Change Application (LCA) is the candidate gate between generation and validation execution. The proposal  $\tilde{\mathbf{c}}_t = \Phi(\mathbf{h}_t, \mathcal{G}_t, \mathcal{B}_t)$  is applied to configured mutable notebook regions and checked against a configured rule set  $\mathcal{C}_1$ . The implementation checks required interfaces and outputs, tensor-shape and numerical failures observed at runtime, and metric-field completeness. Within a fixed repair budget, failed candidates may be locally repaired and checked again; candidates that still fail are recorded as  $\perp$  and excluded from validation selection.

We write the eligible set as  $\Omega_{\text{const}}(\mathcal{D}_{\text{spec}}) = \{\mathbf{c} \in \mathcal{I} : \chi_{\mathbf{I}}(\mathbf{c}, \mathcal{D}_{\text{spec}}) = 1\}$ , where  $\chi_{\mathbf{I}}$  summarizes the configured checks. Operationally,

$$\mathbf{c}_t = \Gamma_{\mathcal{D}}(\tilde{\mathbf{c}}_t; \mathbf{h}_t, \mathcal{G}_t, \mathcal{B}_t, R) \in \Omega_{\text{const}} \cup \{\perp\}. \quad (2)$$

Thus,  $\Gamma_{\mathcal{D}}$  covers code application, checking, bounded repair, and rejection. Passing the implemented checks establishes operational eligibility. Candidate context and outcomes are appended to  $\mathcal{B}_{t+1}$  for subsequent proposals.

### 3.4 PDR: PERFORMANCE-DISCREPANCY REFINEMENT

Performance-Discrepancy Refinement (PDR) converts execution outcomes into diagnostic-guided model-design revisions. After executing an eligible implementation  $\mathbf{c}_t$  under the validation protocol, CellScientist records feedback as

$$\begin{aligned}\mathbf{m}_t &= \mathcal{F}(\mathbf{c}_t; \mathcal{D}_{\text{val}}, \mathcal{E}) \\ &= [\psi_1(\mathbf{c}_t), \dots, \psi_{K_t}(\mathbf{c}_t)], \\ \mathbf{h}_{t+1} &= \mathbf{h}_t \oplus \mathcal{T}_{\text{edit}}(a_t \mid \mathbf{m}_t).\end{aligned}\tag{3}$$

Each diagnostic  $\psi_k$  extracts a signal from execution, validation, or rule checking. PDR maps the diagnostic to an address  $a_t = \mathcal{A}(\mathbf{m}_t, \mathcal{G}_t, \mathcal{B}_t)$  identifying one or more model components;  $\mathcal{T}_{\text{edit}}$  constructs the next revision request. This prompt-conditioned search heuristic uses the recorded diagnostic, task-state context, run history, and LLM-produced focus. Subsequent validation determines candidate retention under the configured selection rule.

### 3.5 TRAJECTORY SELECTION AND WORKFLOW

CellScientist stores each attempt in the run history. Let  $\eta_t = \mathbb{I}\{\mathbf{c}_t \in \Omega_{\text{const}}(\mathcal{D}_{\text{spec}})\}$  indicate whether the proposed implementation passed the configured eligibility checks, and let  $\mathcal{B}_{t+1} = \mathcal{B}_t \cup \{(\mathbf{h}_t, \tilde{\mathbf{c}}_t, \mathbf{c}_t, \mathbf{m}_t, a_t, \eta_t)\}$ . The validation-selected candidate and its design state are given by

$$(\mathbf{h}_{\text{best}}, \mathbf{c}_{\text{best}}) = \text{Select}_{\text{val}}(\{(\mathbf{h}_i, \mathbf{c}_i, \mathbf{m}_i) : \eta_i = 1\}_{i=0}^{T-1}, \mathcal{E}),\tag{4}$$

which returns the best eligible model-design state and executable candidate under the configured validation rule.

## 4 EXPERIMENTS

We evaluate CellScientist at two levels. A high-freedom workflow assesses predictive utility: ‘‘CS-model’’ denotes the executable candidate selected separately for each task, with results summarized over five benchmark folds. These CS-model entries are five-fold validation summaries; they are not presented as a separate final held-out test estimate. A controlled audit fixes the candidate language, low-rank predictor, fitting procedure, data roles, evaluator, LLM settings, initial state, and candidate-evaluation budget to isolate controller behavior on a common reporting fold over five paired seeds. Both use an experiment-specific fixed task contract, persistent history, constrained revision, and validation-based update, while serving distinct experimental purposes.

The experiments address two questions: **RQ1: Predictive utility of selected candidates.** How do the task-specific predictors selected by CellScientist compare with the listed baselines under common data and evaluation protocols? **RQ2: Controlled framework audit.** Under matched candidate language, predictor, data roles, evaluator, LLM settings, initialization, and evaluation budget, does CellScientist improve the held-out global PCC refinement frontier over fixed- $h_0$ , flat-retry, random-routing, and AIDE-style policies, and what operational roles do HRT, LCA, and PDR play?

### 4.1 EXPERIMENTAL SETUP

We evaluate three settings: Cell Painting morphology prediction, LINCS2020 transcriptomic perturbation prediction, and single-cell perturbation prediction. Dataset statistics, baselines, metrics, and implementation details are provided in Appendices C.1, C.2, C.3, and C.4, respectively.

**Cell Painting morphology prediction.** We evaluate three public high-content screening datasets in CellProfiler feature space (Stirling et al., 2021): BBBC036 and BBBC047 (Bray et al., 2017), and CPG0016 (Chandrasekaran et al., 2023; Arevalo et al., 2024). All three morphology benchmarks are evaluated under paired SMILES-based and plate-based splits. Baselines for the main morphology table include TabR (Gorishniy et al., 2024), RealMLP and RF-TD (Holzmüller et al., 2024), FLAML (Wang et al., 2021), Random Search (Gijssbers et al., 2024), and CellForge (Tang et al., 2025). The main text

Table 1: **Core morphology-response benchmarks.** We compare **CS-model**, the task-specific executable model selected by CellScientist, with fixed predictors, search-based baselines, and an agent-designed method on BBBC036, BBBC047, and CPG0016. Results are mean $\pm$ sample standard deviation over five benchmark folds under SMILES-based and plate-based splits using global MSE, PCC, and  $R^2$ . Methods use the reported split, target, and metric definitions, while preprocessing, search procedures, model families, and compute may differ. The fixed-predictor controller audit in Tables 3 and 21 is reported separately. Best mean values among the listed methods are **bolded**.

| Method                       | Venue      | SMILES-based Split       |                          |                          | Plate-based Split        |                          |                          |
|------------------------------|------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
|                              |            | MSE↓                     | PCC↑                     | R <sup>2</sup> ↑         | MSE↓                     | PCC↑                     | R <sup>2</sup> ↑         |
| BBBC036                      |            |                          |                          |                          |                          |                          |                          |
| # Existing reference methods |            |                          |                          |                          |                          |                          |                          |
| TabR                         | ICLR'24    | 3.5518 $\pm$ 0.24        | 0.2747 $\pm$ 0.03        | 0.0119 $\pm$ 0.02        | 2.8759 $\pm$ 0.13        | 0.5065 $\pm$ 0.03        | 0.1855 $\pm$ 0.02        |
| RealMLP                      | NeurIPS'24 | 3.5305 $\pm$ 0.16        | 0.3301 $\pm$ 0.01        | 0.0049 $\pm$ 0.02        | 2.5740 $\pm$ 0.16        | 0.5856 $\pm$ 0.02        | 0.2530 $\pm$ 0.03        |
| RF-TD                        | NeurIPS'24 | 3.5951 $\pm$ 0.19        | 0.2447 $\pm$ 0.01        | 0.0035 $\pm$ 0.01        | 3.5978 $\pm$ 0.14        | 0.2466 $\pm$ 0.02        | 0.0152 $\pm$ 0.01        |
| □ Search-based baselines     |            |                          |                          |                          |                          |                          |                          |
| FLAML                        | MLSys'21   | 3.4217 $\pm$ 0.18        | 0.3486 $\pm$ 0.02        | 0.0379 $\pm$ 0.02        | 2.4918 $\pm$ 0.14        | 0.5987 $\pm$ 0.02        | 0.2784 $\pm$ 0.03        |
| Random Search                | JMLR'24    | 3.4763 $\pm$ 0.21        | 0.3392 $\pm$ 0.02        | 0.0246 $\pm$ 0.02        | 2.5335 $\pm$ 0.15        | 0.5904 $\pm$ 0.02        | 0.2649 $\pm$ 0.03        |
| △ Agent-designed methods     |            |                          |                          |                          |                          |                          |                          |
| CellForge                    | arXiv'25   | 3.4568 $\pm$ 0.19        | 0.3415 $\pm$ 0.03        | 0.0464 $\pm$ 0.02        | 2.5187 $\pm$ 0.14        | 0.6013 $\pm$ 0.02        | 0.2878 $\pm$ 0.03        |
| CS-model                     | Ours       | <b>3.3115</b> $\pm$ 0.20 | <b>0.3688</b> $\pm$ 0.03 | <b>0.0717</b> $\pm$ 0.02 | <b>2.3662</b> $\pm$ 0.15 | <b>0.6212</b> $\pm$ 0.02 | <b>0.3165</b> $\pm$ 0.03 |
| BBBC047                      |            |                          |                          |                          |                          |                          |                          |
| # Existing reference methods |            |                          |                          |                          |                          |                          |                          |
| TabR                         | ICLR'24    | 2.9382 $\pm$ 0.04        | 0.3552 $\pm$ 0.01        | 0.0059 $\pm$ 0.01        | 2.7930 $\pm$ 0.14        | 0.4094 $\pm$ 0.01        | 0.0568 $\pm$ 0.03        |
| RealMLP                      | NeurIPS'24 | 2.8441 $\pm$ 0.03        | 0.3985 $\pm$ 0.01        | 0.0285 $\pm$ 0.01        | 2.6475 $\pm$ 0.10        | 0.4597 $\pm$ 0.01        | 0.1332 $\pm$ 0.02        |
| RF-TD                        | NeurIPS'24 | 2.8850 $\pm$ 0.05        | 0.3746 $\pm$ 0.01        | 0.0281 $\pm$ 0.01        | 2.8174 $\pm$ 0.16        | 0.4005 $\pm$ 0.01        | 0.0615 $\pm$ 0.01        |
| □ Search-based baselines     |            |                          |                          |                          |                          |                          |                          |
| FLAML                        | MLSys'21   | 2.8126 $\pm$ 0.06        | 0.4127 $\pm$ 0.01        | 0.0435 $\pm$ 0.01        | 2.6158 $\pm$ 0.11        | 0.4698 $\pm$ 0.01        | 0.1407 $\pm$ 0.02        |
| Random Search                | JMLR'24    | 2.8314 $\pm$ 0.07        | 0.4059 $\pm$ 0.01        | 0.0358 $\pm$ 0.01        | 2.6342 $\pm$ 0.12        | 0.4635 $\pm$ 0.01        | 0.1362 $\pm$ 0.02        |
| △ Agent-designed methods     |            |                          |                          |                          |                          |                          |                          |
| CellForge                    | arXiv'25   | 2.7869 $\pm$ 0.09        | 0.4154 $\pm$ 0.01        | 0.0476 $\pm$ 0.01        | 2.5908 $\pm$ 0.10        | 0.4772 $\pm$ 0.02        | 0.1471 $\pm$ 0.02        |
| CS-model                     | Ours       | <b>2.7612</b> $\pm$ 0.11 | <b>0.4368</b> $\pm$ 0.01 | <b>0.0687</b> $\pm$ 0.01 | <b>2.5704</b> $\pm$ 0.11 | <b>0.4844</b> $\pm$ 0.02 | <b>0.1538</b> $\pm$ 0.03 |
| CPG0016                      |            |                          |                          |                          |                          |                          |                          |
| # Existing reference methods |            |                          |                          |                          |                          |                          |                          |
| TabR                         | ICLR'24    | 2.2622 $\pm$ 0.12        | 0.0770 $\pm$ 0.01        | -0.5516 $\pm$ 0.04       | 2.0667 $\pm$ 0.04        | 0.1953 $\pm$ 0.01        | -0.4128 $\pm$ 0.02       |
| RealMLP                      | NeurIPS'24 | 1.4174 $\pm$ 0.08        | 0.2616 $\pm$ 0.02        | 0.0297 $\pm$ 0.01        | 1.2105 $\pm$ 0.01        | 0.4418 $\pm$ 0.01        | 0.1717 $\pm$ 0.01        |
| RF-TD                        | NeurIPS'24 | 1.4673 $\pm$ 0.09        | 0.1354 $\pm$ 0.01        | -0.0035 $\pm$ 0.01       | 1.3400 $\pm$ 0.01        | 0.3213 $\pm$ 0.01        | 0.0849 $\pm$ 0.01        |
| □ Search-based baselines     |            |                          |                          |                          |                          |                          |                          |
| FLAML                        | MLSys'21   | 1.2865 $\pm$ 0.05        | 0.3748 $\pm$ 0.01        | 0.1183 $\pm$ 0.01        | 1.1206 $\pm$ 0.03        | 0.5037 $\pm$ 0.02        | 0.2331 $\pm$ 0.02        |
| Random Search                | JMLR'24    | 1.3428 $\pm$ 0.06        | 0.3189 $\pm$ 0.02        | 0.0747 $\pm$ 0.02        | 1.1674 $\pm$ 0.03        | 0.4668 $\pm$ 0.02        | 0.1994 $\pm$ 0.02        |
| △ Agent-designed methods     |            |                          |                          |                          |                          |                          |                          |
| CellForge                    | arXiv'25   | 1.2849 $\pm$ 0.03        | 0.3716 $\pm$ 0.01        | 0.1272 $\pm$ 0.01        | 1.0458 $\pm$ 0.02        | 0.5187 $\pm$ 0.03        | 0.2376 $\pm$ 0.03        |
| CS-model                     | Ours       | <b>1.1113</b> $\pm$ 0.01 | <b>0.5093</b> $\pm$ 0.01 | <b>0.2383</b> $\pm$ 0.01 | <b>0.9937</b> $\pm$ 0.02 | <b>0.5761</b> $\pm$ 0.04 | <b>0.3117</b> $\pm$ 0.04 |

reports global MSE, PCC, and  $R^2$ ; top- $K$  response-feature morphology metrics are provided in Appendix F.2.

**Controlled framework audit.** The audit uses BBBC036/047 under both plate- and SMILES-based settings. All policies share the same initial state, 768-candidate executable language, predictor, fitting procedure, data roles, evaluator, LLM settings, paired seeds, candidate-retention rule, and candidate-evaluation budget. Folds 1–3 fit each candidate, disjoint fold-4 subsets provide controller feedback and candidate selection, and fold 5 is used only for held-out reporting. Each paired seed initializes the same randomized low-rank fitting procedure in every policy, so the reported dispersion includes fitting as well as controller variation without adding noise to evaluated scores. Candidate retention uses the mean of the feedback and selection objectives for every policy. We evaluate five paired seeds at  $B \in \{1, 3, 5, 10\}$ , with global PCC registered as the selection and held-out reporting endpoint. Fixed  $h_0$ , Flat retry, Random routing, Full CellScientist, and an AIDE-style shared-language policy differ in controller policy rather than candidate family or evaluation count.

**Transcriptomic perturbation prediction.** For LINCS2020 L1000 perturbation prediction (Subramanian et al., 2017; Xie et al., 2022), we evaluate two settings: Seven Cell Lines and Full Data. Baselines include DLEPS (Zhu et al., 2021), DeepCE (Pham et al., 2021), CIGER (Pham et al., 2022), MultiDCP (Wu et al., 2022), and TranSiGen (Tong et al., 2024). Metrics include RMSE, PCC, and Positive/Negative P@100.

**Single-cell perturbation prediction.** We evaluate Norman and Papalexi single-cell perturbation benchmarks (Norman et al., 2019; Papalexi et al., 2021), comparing with CellForge (Tang et al., 2025) using

PCC and differential-expression-sensitive  $PCC_{DE}$ . Papalexí RNA and protein are evaluated as separate readouts.

Table 2: **LINCS2020 transcriptomic perturbation prediction.** **CS-model** achieves the best mean results among the listed methods in both the Seven Cell Lines and Full Data settings. Each uses an independent task-specific search. Best means are **bolded**.

| Dataset    | Model           | RMSE ↓                            | PCC ↑                             | Positive<br>P@100 ↑               | Negative<br>P@100 ↑               |
|------------|-----------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| Seven Cell | DLEPS           | 1.4954 $\pm$ 0.04                 | 0.4268 $\pm$ 0.02                 | 0.2460 $\pm$ 0.01                 | 0.3087 $\pm$ 0.01                 |
|            | DeepCE          | 1.7728 $\pm$ 0.03                 | 0.4121 $\pm$ 0.01                 | 0.2282 $\pm$ 0.02                 | 0.2713 $\pm$ 0.02                 |
|            | CIGER           | 4.4646 $\pm$ 1.06                 | 0.4165 $\pm$ 0.01                 | 0.2337 $\pm$ 0.02                 | 0.2849 $\pm$ 0.02                 |
|            | MultiDCP        | 1.7697 $\pm$ 0.03                 | 0.4461 $\pm$ 0.01                 | 0.2378 $\pm$ 0.02                 | 0.2872 $\pm$ 0.01                 |
|            | TranSiGen       | 0.6903 $\pm$ 0.02                 | 0.5332 $\pm$ 0.02                 | 0.3785 $\pm$ 0.01                 | 0.3886 $\pm$ 0.02                 |
|            | <b>CS-model</b> | <b>0.5088<math>\pm</math>0.02</b> | <b>0.6183<math>\pm</math>0.02</b> | <b>0.4450<math>\pm</math>0.02</b> | <b>0.4481<math>\pm</math>0.02</b> |
| Full Data  | TranSiGen       | 0.5186 $\pm$ 0.01                 | 0.6076 $\pm$ 0.01                 | 0.4353 $\pm$ 0.01                 | 0.4403 $\pm$ 0.01                 |
|            | <b>CS-model</b> | <b>0.4967<math>\pm</math>0.01</b> | <b>0.6368<math>\pm</math>0.01</b> | <b>0.4583<math>\pm</math>0.01</b> | <b>0.4652<math>\pm</math>0.01</b> |

#### 4.2 PREDICTIVE UTILITY OF SELECTED CANDIDATES (RQ1)

Predictive evidence comprises morphology benchmarks (Table 1), a descriptive matched-fold comparison (Figure 4), LINCS2020 transcriptomic prediction (Table 2), and single-cell perturbation results (Appendix F.3).

**Core morphology-response benchmarks.** Table 1 compares the task-specific **CS-model** with fixed predictors, search-based baselines, and the agent-designed CellForge workflow. The SMILES-based split holds out exact compound identities; the plate-based split holds out experimental plates. CS-model achieves the best mean MSE, PCC, and  $R^2$  among the listed methods under both splits, supporting predictive utility in the high-freedom workflow. The controlled audits in Tables 3 and 21 fix a low-rank predictor and fitting procedure to attribute differences to controller policy; their absolute PCC values are specific to that protocol. Appendix F.2 reports top- $K$  response-feature metrics. Figure 4 shows fold-level  $R^2$  comparisons under the same five benchmark folds, with paired  $t$ -test markers as descriptive summaries.

**Applicability across response spaces.** Table 2 reports separately selected CS-model candidates for LINCS2020 transcriptomic perturbation prediction. Appendix F.3 reports single-cell results on Norman and the RNA and protein readouts of Papalexí. Each task uses an independent search and training run, supporting workflow applicability across response spaces.

#### 4.3 MATCHED-BUDGET FRAMEWORK AUDIT (RQ2)

The controlled audit fixes candidate freedom to measure the contribution of the CellScientist controller. Policies share  $h_0$ , the 768-candidate executable language, low-rank predictor, fitting procedure, fold roles, diagnostic record, and candidate-evaluation budget. This predictor differs from the task-specific CS-model architectures in Table 1. Global PCC, computed after flattening prediction and observation matrices, is the sole metric for validation-based candidate retention and held-out frontier reporting. The full validation record remains available for diagnostic component attribution.

**Matched policies.** Fixed  $h_0$  performs no revision. Flat retry selects among up to 12 unseen one-step neighbors using validation feedback but without a typed edit address. Random routing samples one currently editable address uniformly, after which the same deterministic atomic-edit instantiation used by CellScientist is applied within that address. The AIDE-style shared-language policy allocates the available budget first to at most five independent drafts and then to greedy improvement when budget remains; in every case, the total number of evaluated candidates is exactly  $B$ . CellScientist ranks typed addresses from the current diagnostics and HRT history, exposes at most two legal address-candidate pairs, and uses Gemini 3 Pro to select one allow-listed pair before LCA checking and execution. Capacity continuation is exposed only after the matched lower-capacity predecessor improves the feedback objective.

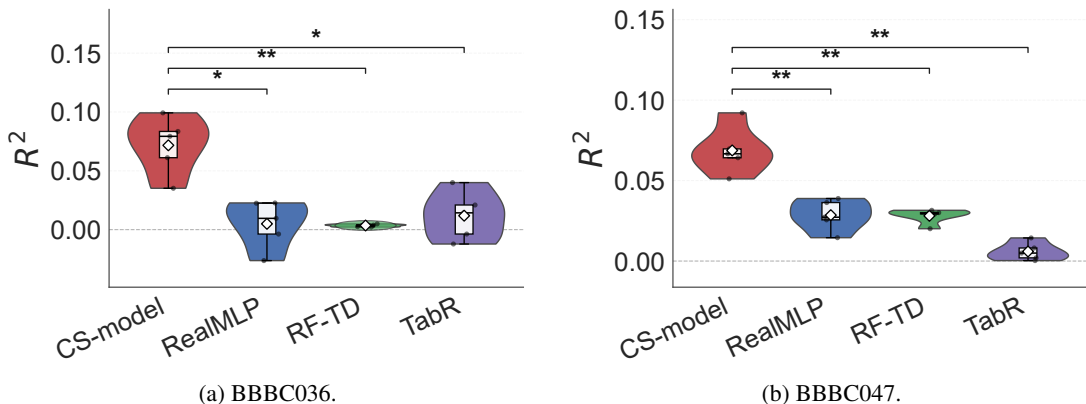


Figure 4: **Fold-level comparison on BBBC036 and BBBC047.** Five-fold  $R^2$  distributions compare **CS-model** with reference predictors. Violins show distributions; boxes, points, and diamonds denote quartiles, folds, and means. Stars give uncorrected paired  $t$ -tests:  $*p < 0.05$ ,  $**p < 0.01$ ,  $***p < 0.001$ .

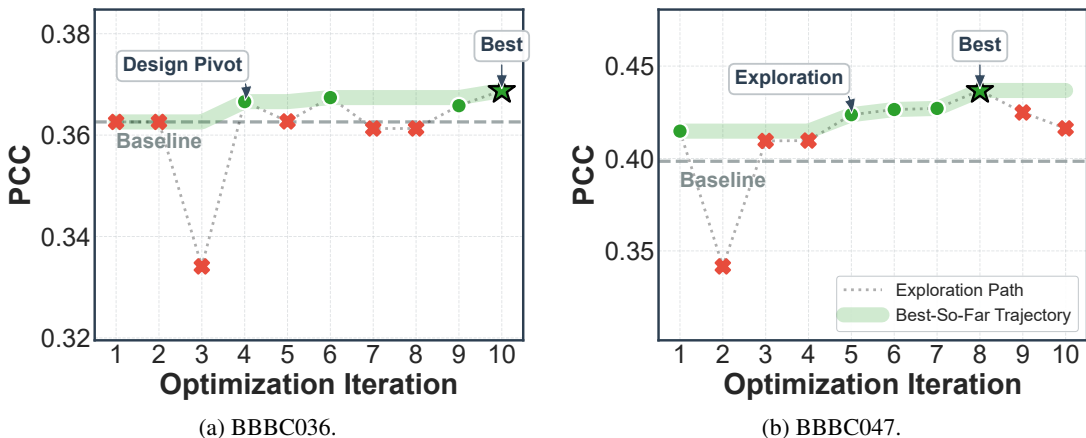


Figure 5: **High-freedom workflow trajectories.** Raw validation utilities and retained best-so-far trajectories over ten refinement steps under the SMILES-based split. Exploratory regressions remain visible alongside the validation-selected path.

**Controller attribution.** Table 3 exposes the absolute task-level results rather than only aggregate differences. Across the four paired settings, CellScientist exceeds Fixed  $h_0$ , Flat retry, Random routing, and AIDE-style search by 0.0939, 0.0837, 0.0361, and 0.0368 PCC at  $B = 3$ , respectively; all four paired-setting 95% CIs are above zero. At  $B = 5$ , the corresponding effects are 0.0984, 0.0648, 0.0373, and 0.0215, again with CIs above zero. At  $B = 10$ , CellScientist has higher mean PCC than Random routing in three settings and a similar mean on BBBC036–SMILES (0.2346 versus 0.2347), while retaining a 0.0181 advantage over AIDE-style search (95% CI [0.0052, 0.0309]). The normalized full-frontier AUC improvements over Fixed  $h_0$ , Flat retry, Random routing, and AIDE-style search are 0.0877, 0.0608, 0.0237, and 0.0215, with respective 95% CIs of [0.0680, 0.1074], [0.0414, 0.0802], [0.0172, 0.0303], and [0.0117, 0.0314].

Appendix D (Table 21) reports the absolute backbone-specific results underlying the aggregate routing effects. With Gemini 3 Pro, CellScientist exceeds Random routing by 0.0361 at  $B = 3$  and 0.0373 at  $B = 5$ ; the normalized frontier-AUC difference is 0.0237 (95% CI [0.0172, 0.0303]). Qwen2.5-5B-Instruct reproduces these effects at 0.0364, 0.0341, and 0.0217 (95% CI [0.0158, 0.0275]), respectively. Similar effects across the four task/split settings support the controller workflow across these two backbones within the controlled audit.

**HRT mechanism.** Table 4a removes only persistent proposal history while fixing the trained-candidate budget. HRT eliminates repeated proposals and more than doubles the unique-proposal ratio. Under HRT w/o History, repeated IDs are recorded and replaced by legal unseen candidates before evaluation,

Table 3: **Five-seed matched-budget controller comparison on held-out global PCC.** The upper panel reports mean $\pm$ sample standard deviation over five paired controller and fitting seeds. The lower panel reports CellScientist-minus-comparator effects with paired-setting 95%  $t$ -confidence intervals across the four task/split settings, computed from unrounded run-level values; confidence intervals are shown in smaller type. All policies use the same initial candidate, executable candidate language, data roles, fitting procedure, and number of evaluated candidates. The uninformative common- $h_0$  row at  $B = 1$  is omitted. Best means at each task, split, and budget are **bolded**.

| Dataset                                                         | Split  | Budget | Fixed $h_0$               | Flat retry                | Random routing             | AIDE-style                | CellScientist              |
|-----------------------------------------------------------------|--------|--------|---------------------------|---------------------------|----------------------------|---------------------------|----------------------------|
| BBBC036                                                         | Plate  | 3      | 0.1664 $\pm$ 0.0005       | 0.1828 $\pm$ 0.0005       | 0.2250 $\pm$ 0.0493        | 0.2244 $\pm$ 0.0276       | <b>0.2650</b> $\pm$ 0.0005 |
|                                                                 |        | 5      | 0.1664 $\pm$ 0.0005       | 0.2073 $\pm$ 0.0056       | 0.2277 $\pm$ 0.0502        | 0.2477 $\pm$ 0.0294       | <b>0.2674</b> $\pm$ 0.0003 |
|                                                                 |        | 10     | 0.1664 $\pm$ 0.0005       | 0.2175 $\pm$ 0.0061       | 0.2626 $\pm$ 0.0077        | 0.2562 $\pm$ 0.0205       | <b>0.2693</b> $\pm$ 0.0012 |
| BBBC036                                                         | SMILES | 3      | 0.1547 $\pm$ 0.0005       | 0.1583 $\pm$ 0.0046       | 0.2052 $\pm$ 0.0413        | 0.2058 $\pm$ 0.0230       | <b>0.2336</b> $\pm$ 0.0016 |
|                                                                 |        | 5      | 0.1547 $\pm$ 0.0005       | 0.1882 $\pm$ 0.0129       | 0.2049 $\pm$ 0.0417        | 0.2231 $\pm$ 0.0169       | <b>0.2346</b> $\pm$ 0.0009 |
|                                                                 |        | 10     | 0.1547 $\pm$ 0.0005       | 0.1948 $\pm$ 0.0005       | <b>0.2347</b> $\pm$ 0.0016 | 0.2235 $\pm$ 0.0164       | 0.2346 $\pm$ 0.0009        |
| BBBC047                                                         | Plate  | 3      | 0.2442 $\pm$ 0.0008       | 0.2500 $\pm$ 0.0055       | 0.3032 $\pm$ 0.0500        | 0.3013 $\pm$ 0.0381       | <b>0.3396</b> $\pm$ 0.0004 |
|                                                                 |        | 5      | 0.2442 $\pm$ 0.0008       | 0.2630 $\pm$ 0.0130       | 0.3066 $\pm$ 0.0514        | 0.3153 $\pm$ 0.0399       | <b>0.3457</b> $\pm$ 0.0004 |
|                                                                 |        | 10     | 0.2442 $\pm$ 0.0008       | 0.2777 $\pm$ 0.0005       | 0.3458 $\pm$ 0.0027        | 0.3191 $\pm$ 0.0372       | <b>0.3481</b> $\pm$ 0.0009 |
| BBBC047                                                         | SMILES | 3      | 0.2504 $\pm$ 0.0004       | 0.2651 $\pm$ 0.0002       | 0.3131 $\pm$ 0.0536        | 0.3123 $\pm$ 0.0346       | <b>0.3529</b> $\pm$ 0.0010 |
|                                                                 |        | 5      | 0.2504 $\pm$ 0.0004       | 0.2916 $\pm$ 0.0003       | 0.3207 $\pm$ 0.0582        | 0.3371 $\pm$ 0.0417       | <b>0.3614</b> $\pm$ 0.0003 |
|                                                                 |        | 10     | 0.2504 $\pm$ 0.0004       | 0.2920 $\pm$ 0.0003       | 0.3668 $\pm$ 0.0073        | 0.3560 $\pm$ 0.0359       | <b>0.3751</b> $\pm$ 0.0017 |
| CellScientist advantage: mean effect with paired-setting 95% CI |        |        |                           |                           |                            |                           |                            |
| Budget                                                          |        |        | vs. Fixed $h_0$           | vs. Flat                  | vs. Random                 | vs. AIDE-style            |                            |
| $B = 3$                                                         |        |        | +0.0939 [+0.0773,+0.1104] | +0.0837 [+0.0734,+0.0940] | +0.0361 [+0.0275,+0.0448]  | +0.0368 [+0.0270,+0.0466] |                            |
| $B = 5$                                                         |        |        | +0.0984 [+0.0774,+0.1193] | +0.0648 [+0.0404,+0.0892] | +0.0373 [+0.0292,+0.0454]  | +0.0215 [+0.0088,+0.0342] |                            |
| $B = 10$                                                        |        |        | +0.1029 [+0.0737,+0.1320] | +0.0612 [+0.0306,+0.0919] | +0.0043 [-0.0018,+0.0104]  | +0.0181 [+0.0052,+0.0309] |                            |

preserving the matched budget. The ablation measures the additional unique-proposal coverage provided by persistent history.

**LCA and PDR mechanisms.** Table 4, panels b and c, summarizes the registered LCA and held-out PDR audits. LCA correctly classifies all 60 registered cases and contains all 40 violations. Structured PDR localizes and repairs all 15 held-out faults, with no unrelated edits and one repair round on average. Appendix E provides outcome-level and component-address breakdowns. These audits complement the predictive frontier in Table 3 with direct measurements of operational behavior.

Table 4: **Compact HRT, LCA, and PDR mechanism audits.** (a) HRT compares 20 matched Gemini-controlled runs with 200 candidate evaluations; HRT and w/o History use 170 and 180 proposal steps. HRT w/o History removes only the persistent seen/failed record while retaining the current state, diagnostics, candidate language, and controller. (b) LCA summarizes 40 violations and 20 legal controls; it makes 60/60 correct decisions, contains 40/40 violations, and avoids 30 invalid evaluations. (c) PDR compares Flat retry, Random routing, and Structured PDR over 15 held-out faults. Full LCA and PDR breakdowns appear in Appendix E; best values are **bolded**.

| (a) HRT history.                |               |           | (b) LCA outcomes. |       |                | (c) PDR held-out results. |             |             |             |             |
|---------------------------------|---------------|-----------|-------------------|-------|----------------|---------------------------|-------------|-------------|-------------|-------------|
| Measure                         | HRT           | w/o Hist. | Outcome           | Cases | Result         | Policy                    | Top-1       | Repair      | Edits       | Rounds      |
| Unique proposals $\uparrow$     | <b>170</b>    | 82        | Repair            | 15    | 15/15 repaired | Flat                      | 13.3%       | 33.3%       | 3.20        | 4.13        |
| Repeated proposals $\downarrow$ | <b>0</b>      | 98        | Pre-run block     | 15    | 15/15 blocked  | Random                    | 6.7%        | 60.0%       | 3.27        | 3.87        |
| Repeat rate $\downarrow$        | <b>0.0%</b>   | 54.4%     | Post-run block    | 10    | 10/10 blocked  | <b>Structured</b>         | <b>100%</b> | <b>100%</b> | <b>0.00</b> | <b>1.00</b> |
| Unique ratio $\uparrow$         | <b>100.0%</b> | 45.6%     | Legal controls    | 20    | 20/20 accepted |                           |             |             |             |             |

## 5 CONCLUSION

CellScientist converts execution and validation feedback into local model revisions under a fixed task contract, retaining an auditable history of successful and unsuccessful attempts. Task-specific benchmarks show predictive utility across cellular response spaces, while the controlled audit isolates the contribution of feedback-guided revision under matched conditions. Operational audits connect history tracking, contract checks, and discrepancy routing to measured outcomes. The workflow produces both a validation-selected model and an inspectable account of its development. Its scope remains bounded by the controller, prompts, candidate language, and initial hypotheses; future work should examine more robust routing and transferable revision knowledge.

## REPRODUCIBILITY STATEMENT

The operational algorithm and notation are provided in Appendix A. Dataset definitions, baseline protocols, evaluation metrics, implementation settings, and controller configurations are detailed in Appendices C.1–C.4. The case studies retain failed and regressive candidates alongside validation-selected outcomes, while the registered LCA and held-out PDR audits are documented in Appendix E.

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## A ALGORITHM AND NOTATION

**Coupled model-design revision.** Perturbation-response models combine interdependent choices in perturbation representation, cellular-context encoding, dose conditioning, feature interaction, objective design, and optimization. Changing one component can alter the behavior of others. Model revision therefore requires associating observed outcomes with candidate explanations and possible edit targets. An explicit record of component dependencies, accumulated evidence, and edit scope supports this process without establishing causal attribution.

**Hypothesis–implementation alignment.** Evaluating a model-design hypothesis requires an executable implementation that preserves the task’s data, prediction target, interface, and evaluation semantics. Execution and validation outcomes then inform which design assumptions and components merit reconsideration. We formulate this as a *model-design revision-routing problem*: mapping observed discrepancies to local design revisions while maintaining a fixed task contract.

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### Algorithm 1: CellScientist Workflow

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**Require:** Fixed task specification  $\mathcal{D}_{\text{spec}}$ , initial model-design state  $\mathbf{h}_0$ , task-state graph  $\mathcal{G}_0$ , iteration budget  $T$ , repair budget  $R$

- 1: Initialize run history  $\mathcal{B}_0 \leftarrow \emptyset$  and best eligible state  $(\mathbf{h}_{\text{best}}, \mathbf{c}_{\text{best}}) \leftarrow (\mathbf{h}_0, \perp)$
  - 2: **for**  $t = 0$  to  $T - 1$  **do**
  - 3:   Summarize the current model-design state, task-state graph, and run history
  - 4:   Request a focus area and candidate edits  $\tilde{\mathbf{c}}_t \leftarrow \Phi(\mathbf{h}_t, \mathcal{G}_t, \mathcal{B}_t)$
  - 5:   Apply edits and configured checks with at most  $R$  repair attempts:  $\mathbf{c}_t \leftarrow \Gamma_{\mathcal{D}}(\tilde{\mathbf{c}}_t; \mathbf{h}_t, \mathcal{G}_t, \mathcal{B}_t, R)$
  - 6:   **if**  $\mathbf{c}_t \neq \perp$  **then**
  - 7:     Execute  $\mathbf{c}_t$  under the validation view of  $\mathcal{D}_{\text{spec}}$
  - 8:     Record execution status, configured rule checks, and validation feedback  $\mathbf{m}_t$
  - 9:     Set  $\eta_t \leftarrow 1$
  - 10:   **else**
  - 11:     Record execution or required-output failure in  $\mathbf{m}_t$  and set  $\eta_t \leftarrow 0$
  - 12:   **end if**
  - 13:   Route the diagnostic to an edit address  $a_t \leftarrow \mathcal{A}(\mathbf{m}_t, \mathcal{G}_t, \mathcal{B}_t)$
  - 14:   Form the next design record  $\mathbf{h}_{t+1} \leftarrow \mathbf{h}_t \oplus \mathcal{T}_{\text{edit}}(a_t \mid \mathbf{m}_t)$
  - 15:   Append the addressed evidence and status to obtain  $\mathcal{G}_{t+1}$
  - 16:   Update run history  $\mathcal{B}_{t+1} \leftarrow \mathcal{B}_t \cup \{(\mathbf{h}_t, \tilde{\mathbf{c}}_t, \mathbf{c}_t, \mathbf{m}_t, a_t, \eta_t)\}$
  - 17:   **if**  $\eta_t = 1$  and  $\mathbf{c}_t$  improves the configured validation score **then**
  - 18:      $(\mathbf{h}_{\text{best}}, \mathbf{c}_{\text{best}}) \leftarrow (\mathbf{h}_t, \mathbf{c}_t)$
  - 19:   **end if**
  - 20: **end for**
  - 21: **return**  $(\mathbf{h}_{\text{best}}, \mathbf{c}_{\text{best}})$  and the recorded trajectory  $(\mathcal{G}_T, \mathcal{B}_T)$
-

Table 5: Notation used in CellScientist.

| Symbol                                                                            | Description                                                                                         |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| $\mathcal{H}$                                                                     | Model-design space containing task-specific representation, fusion, objective, and training choices |
| $\mathcal{I}$                                                                     | Space of executable candidate programs                                                              |
| $\mathcal{D}_{\text{spec}}$                                                       | Fixed task specification $(\mathcal{X}, \mathcal{P}, \mathcal{Y}, \mathcal{S}, \mathcal{E})$        |
| $\mathcal{X}, \mathcal{P}, \mathcal{Y}$                                           | Cellular inputs, perturbation descriptors, and response targets                                     |
| $\mathcal{S}$                                                                     | Data-partition protocol used by the configured task                                                 |
| $\mathcal{E}$                                                                     | Evaluation functional used for candidate scoring and selection                                      |
| $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{test}}$ | Training, controller-feedback/selection, and untouched reporting views induced by $\mathcal{S}$     |
| $\mathbf{h}_t$                                                                    | Model-design record at iteration $t$                                                                |
| $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t)$                                  | Structured task-state graph used for routing context and audit                                      |
| $\mathcal{V}_t, \mathcal{E}_t$                                                    | Task nodes and directed dependency edges                                                            |
| $\mathcal{B}_t$                                                                   | Run history storing attempted designs, implementations, and outcomes                                |
| $\mathcal{Z}_t = (\mathbf{h}_t, \mathcal{G}_t, \mathcal{B}_t)$                    | Complete controller state at iteration $t$                                                          |
| $T, R$                                                                            | Candidate-iteration budget and bounded LCA repair budget                                            |
| $\Phi$                                                                            | LLM-conditioned candidate-proposal operator                                                         |
| $\Gamma_{\mathcal{D}}$                                                            | Code-application, configured-check, bounded-repair, and rejection procedure                         |
| $\tilde{\mathbf{c}}_t$                                                            | Proposed candidate edits before execution checks                                                    |
| $\mathbf{c}_t$                                                                    | Candidate produced after code application and checks                                                |
| $\perp$                                                                           | Candidate that failed configured execution or required-output checks                                |
| $\eta_t$                                                                          | Indicator that $\mathbf{c}_t$ passed the configured eligibility checks                              |
| $\Omega_{\text{const}}$                                                           | Candidates that pass the implemented checks under $\mathcal{D}_{\text{spec}}$                       |
| $\chi_{\mathbf{I}}, \mathcal{C}_{\mathbf{I}}$                                     | Eligibility indicator and its configured implementation-rule set                                    |
| $\mathcal{F}$                                                                     | Feedback extraction operator                                                                        |
| $\mathbf{m}_t$                                                                    | Structured feedback vector at iteration $t$                                                         |
| $\psi_k$                                                                          | Individual diagnostic signal used to construct $\mathbf{m}_t$                                       |
| $K_t$                                                                             | Number of diagnostic signals used at iteration $t$                                                  |
| $\mathcal{A}, a_t$                                                                | Diagnostic-to-task association procedure and returned task address                                  |
| $\mathcal{T}_{\text{edit}}$                                                       | Revision-request operator conditioned on recorded feedback                                          |
| $\oplus$                                                                          | Application of a revision to a model-design record                                                  |
| $\text{Select}_{\text{val}}$                                                      | Selection rule over eligible validation runs                                                        |
| $(\mathbf{h}_{\text{best}}, \mathbf{c}_{\text{best}})$                            | Validation-selected design record and executable candidate                                          |

## B DETAILED CASE STUDIES OF AUDITABLE MODEL-REVISION TRAJECTORIES

We reconstruct two model-revision trajectories on morphology-based perturbation-response tasks. Each record preserves execution errors, rule checks, validation outcomes, rejected candidates, and retained candidates under a fixed evaluation protocol. Phase 1–3 follow the pipeline logs: diagnosis and task setup, initial candidate generation, and subsequent review. HRT, LCA, and PDR operate throughout these phases.

On BBBC036, CellScientist retained a Drug–Dose Multiplicative Interaction Architecture (DD-MIA) candidate after evaluating several representation and fusion variants. The selected design combines graph-based compound encoding, batch-related regularization, and dose-aware multiplicative fusion. The trajectory supports selecting this combination under the task-specific validation protocol.

On BBBC047, modulation and VAE-style candidates showed both gains and regressions. The trajectory recorded possible underuse of the supplied control-reference profile as a working diagnostic. CellScientist subsequently retained Ref-RGAF, which adds a deterministic control-reference encoding to the decoder. The record links this diagnostic to a local revision and its validation outcome.

These cases make CellScientist’s selection process inspectable: the records identify attempted assumptions, failed or regressive branches, and validation-selected executable candidates. They complement the quantitative results by tracing local revisions and selection decisions under a fixed protocol.

## B.1 DETAILED CASE STUDY A

### Case A: Auditable Revision Toward DD-MIA

*Trajectory including negative results, crashes, regressions, and retained candidates.*

#### Case Study Overview

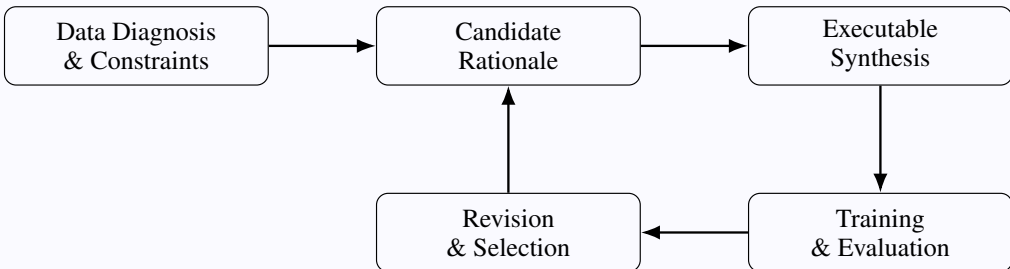
This section reconstructs the model-revision trajectory of **CellScientist** on BBBC036 under the SMILES-based split. The trajectory moves from simple feature-based modeling toward **DD-MIA**, a candidate architecture that separates compound-structure representation from dose-dependent response scaling.

**What this case study illustrates.** The trace records how CellScientist diagnosed data and batch-related issues, proposed design rationales, implemented candidates, and retained the highest-scoring eligible candidate under the validation protocol.

**Trajectory evidence.** The trajectory documents how recorded diagnostics and validation outcomes were associated with local design revisions and empirical candidate selection.

#### Closed-Loop Revision Trajectory

**Auditable closed-loop design record:**  
diagnosis → executable candidate → validation feedback



**Interpretation.** The loop records a practical model-design process: *diagnose* → *propose* → *implement* → *test* → *revise*.

**(A) From Representation Choice to Design Rationale** The trajectory is non-monotonic: several candidates underperformed despite valid execution, and other candidates failed during execution. The complete record shows which components were retained, revised, or rejected at each step.

Table 9 summarizes representative scored candidates from the trajectory. Graph-based compound representation was retained, while several more complex fusion variants regressed or failed.

#### Key Observation

The trajectory preserves both validation-retained candidates and negative evidence. Several higher-capacity variants regressed, while the validation rule retained a more structured candidate. This record makes the empirical selection process auditable.

**(B) Auditable Model-Revision Trajectory** We report the diagnosis and revision phases, including failures and regressions. These records show how the validation rule selected the retained candidate. **Phase 1: Problem Formulation and Data Diagnosis.** Before architecture revision, the system defines the prediction target and records operational constraints. Table 6 summarizes the data checks that informed later candidates.

Table 6: Phase 1 Execution Trace: Diagnosis and Constraint Identification.

| Step | Activity                    | Artifact                 | Design / Operational Insight                                                                                                                                                                                                          |
|------|-----------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Data Ingestion & Pairing    | HDF5 Database            | <b>Operational repair:</b> Resolved HDF5 sizing mismatch by enforcing type and shape alignment. Established the $f(\text{ControlRef}, \text{Chem}, \text{Dose}) \rightarrow \text{TreatedResponse}$ mapping used by later candidates. |
| 2    | Statistical Diagnosis       | ANOVA/PCA                | <b>Operational repair:</b> Corrected a syntax error in the statistical constraint formula. <b>Data cue:</b> Observed a measurable plate-related signal in the feature space, motivating later checks for batch-related shortcuts.     |
| 3    | Dose-Response Profiling     | Volcano Plot             | <b>Response cue:</b> Identified features with strong perturbation response and observed non-linear, heterogeneous dose behavior. This suggested that a purely linear response model may be insufficient.                              |
| 4    | Technical-Factor Inspection | Assay-Variable Schematic | <b>Constraint definition:</b> The persistence of plate effects after normalization motivated the use of batch-related regularization in later candidates.                                                                             |

## Key Observation

**From diagnosis to design.** ANOVA/PCA diagnostics associated part of the feature variation with technical factors. This recorded observation motivated batch-related regularization in later candidates.

**Phase 2: Generate and Execute.** Phase 2 evaluated candidate representations and recorded setup errors and unstable training (Table 7). The highest-scoring eligible candidate used graph-based molecular encoding and exceeded the initial flat-vector candidate in validation PCC. Subsequent designs retained this representation based on the observed comparison. The Phase 2 baseline is an exploratory setup-trace fit; it is not the reference used for the Phase 3 aggregate gain.

Table 7: Phase 2 Execution Trace: Structural Representation Trial.

| Iter | Strategy                              | Outcome         | PCC           | Design Analysis & Operational Event                                                                                                                                                               |
|------|---------------------------------------|-----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Initial Debug / Baseline <sup>a</sup> | Baseline        | 0.3172        | <b>Operational:</b> Pipeline stabilized through a setup-node repair. <b>Baseline:</b> Established lower-bound performance using standard flat-vector features.                                    |
| 2    | <b>EquiMorph-GAT</b>                  | <b>Retained</b> | <b>0.3660</b> | <b>Design cue:</b> Explicit molecular topology encoding improved over the initial flat-vector candidate in this trace, suggesting that compound-structure representation is useful for this task. |
| 3    | MD_AdvN (Adversarial)                 | Regression      | 0.3378        | <b>Negative result:</b> Adversarial normalization alone obtained a lower validation PCC and was rejected by the validation rule.                                                                  |
| 4    | Innovation Variant                    | Regression      | 0.3656        | <b>Validation check:</b> A graph-based variant produced similar performance across a second implementation attempt.                                                                               |
| 5    | Baseline Re-test                      | Regression      | 0.2827        | <b>Regression:</b> The simplified baseline obtained a lower validation PCC in this trace; subsequent candidates retained the structured representation.                                           |

<sup>a</sup>Exploratory Phase 2 baseline; Phase 3 gains use the separately refit Phase 3 reference shown in Table 9.

### Key Observation

**Representation cue.** The graph-based candidate improved over simplified flat-vector base-lines within this recorded trajectory, providing task-specific validation evidence for retaining compound-structure encoding.

**Phase 3: Review and Revise.** Phase 3 evaluated how to combine the control-reference profile, compound representation, and dose information. Table 8 reports all iterations, including crashes and candidates rejected for lower validation performance.

Table 8: **Phase 3 Full Iteration Trace (Iter 1–10).**

| Iter | Strategy              | Outcome         | PCC           | Design Analysis                                                                                                                               |
|------|-----------------------|-----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Chem-Modulated FiLM   | Regression      | 0.3626        | Candidate assumption: affine modulation is sufficient. Result: validation PCC was lower than the retained graph-based model.                  |
| 2    | Res-EquiMorph + Token | Crash           | N/A           | Candidate assumption: tokenization may improve interaction modeling. Result: instability caused a crash and prevented a valid score.          |
| 3    | Tokenized Adversarial | Regression      | 0.3341        | Candidate assumption: finer granularity would improve robustness. Result: the tokenized variant obtained lower validation PCC.                |
| 4    | DGAN (Gated-Adv)      | Improved        | 0.3666        | Design pivot: adversarial regularization was restricted to the morphology branch, while chemistry was used to gate the perturbation response. |
| 5    | Res-DGAN              | Regression      | 0.3627        | Negative add-on: an additional residual skip produced a lower validation PCC in this run.                                                     |
| 6    | Attn-DGAN (Attention) | Improved        | 0.3674        | Refinement: cross-attention provided a more structured interaction between morphology and compound features.                                  |
| 7    | Dose-Gated Attention  | Regression      | 0.3613        | Negative result: directly gating attention by dose reduced validation PCC in this candidate.                                                  |
| 8    | Dose-Conditioned Res  | Crash           | N/A           | Failure mode: custom dose-conditioned gating introduced shape mismatches and training instability.                                            |
| 9    | Projected-Token Attn  | Improved        | 0.3658        | Partial recovery: identity initialization helped, while the added parameters produced a comparable validation result.                         |
| 10   | <b>DD-MIA</b>         | <b>Retained</b> | <b>0.3688</b> | Selected design: separates compound-structure representation from dose-dependent intensity scaling using multiplicative fusion.               |

### Key Observation

**Why failures matter.** The record preserves crashes and regressions from direct dose injection into attention and more complex tokenized variants. These outcomes expose the sequence of attempted revisions and validation decisions.

**(C) Validation-Selected Design: DD-MIA** The retained **DD-MIA** candidate combines three implementation choices; the trajectory evaluates their combination without isolating each component’s contribution.

**(1) Structural compound representation.** The graph-based representation was retained because it improved validation behavior relative to simpler flat-vector candidates in this trace.

**(2) Batch-related regularization.** Diagnostics suggested technical variation, motivating an adversarial regularization component intended to reduce reliance on plate-related shortcuts in the selected design.

**Failure Mode / Design Risk**

Plate or batch signals can become shortcuts under some split settings. The adversarial component was included to address this risk during candidate training.

**(3) Chemistry–dose multiplicative factorization.** The validation-selected candidate uses the following multiplicative fusion form:

$$\Delta \hat{P} = \text{Chem}(\mathcal{G}) \odot \text{Dose}(d). \quad (5)$$

In this implementation, one learned component is conditioned on compound structure and the other on dose. Validation selected this architectural parameterization within the reported trajectory.

**Design Rationale Encoded as Architecture**

The factorization encodes a task-specific architectural preference in which compound and dose information interact multiplicatively. Its retention follows the validation evidence recorded in this trajectory.

**(D) Evaluation and Trajectory Audit** The retained DD-MIA candidate achieved a five-fold validation aggregate PCC of 0.3688 in this trajectory. This is the same validation aggregate reported for the selected CS-model in Table 1, not a separate final held-out evaluation. The record documents its selection path, including failed candidates, regressions, and retained components under the fixed validation protocol.

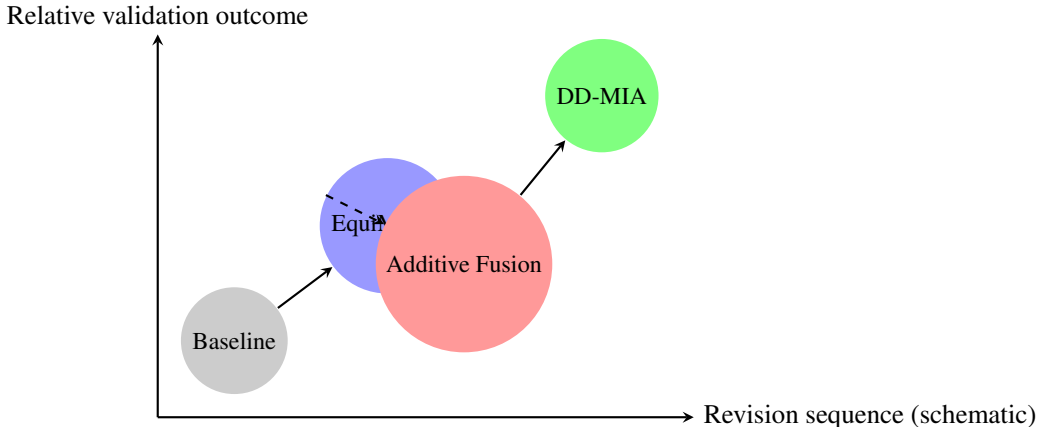


Figure 6: Auditable model-revision trajectory: CellScientist follows a non-monotonic path through candidate architectures, retaining validated components while recording underperforming branches.

Table 9: **Trajectory of Candidate Performance.**

| Activity Step         | Design Rationale   | PCC           | $\Delta$ vs Base | Status             |
|-----------------------|--------------------|---------------|------------------|--------------------|
| Baseline <sup>a</sup> | MLP (Flat Vectors) | 0.3396        | 0.00%            | Ref                |
| Initial Search        | EquiMorph-GAT      | 0.3660        | +7.77%           | Retained candidate |
| Iter 1                | Additive FiLM      | 0.3626        | +6.77%           | Regression         |
| Iter 6                | Attn-DGAN          | 0.3674        | +8.18%           | Improved           |
| Iter 10               | DD-MIA             | <b>0.3688</b> | <b>+8.60%</b>    | Retained candidate |

<sup>a</sup>Phase 3 reference fit, separately refit from the exploratory Phase 2 baseline (0.3172); all Phase 3 gains in this table use 0.3396.

### Key Observation

The trajectory records both validation-retained improvements and negative evidence. DD-MIA is the candidate retained for this benchmark under the reported validation-selection rule.

**(E) Summary** The BBBC036 trajectory connects representation choices, batch-related diagnostics, and fusion revisions to recorded execution and validation outcomes. CellScientist retained DD-MIA under the fixed selection rule while preserving failed and regressive alternatives.

## B.2 DETAILED CASE STUDY B

The introduction summarizes the BBBC047 trajectory in Figure 2. The detailed record below links its working diagnostics, local revisions, validation outcomes, and retained candidate.

### Case B: Auditable Revision Toward Ref-RGAF

*Trajectory including negative results, crashes, regressions, and retained candidates.*

#### Case Study Overview

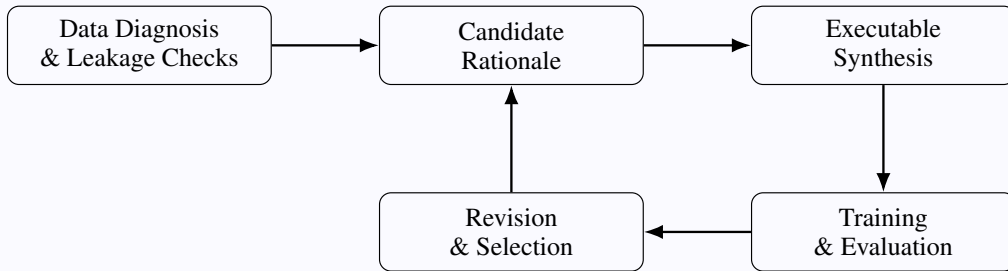
This section reconstructs the model-revision trajectory of **CellScientist** on BBBC047 under the SMILES-based split. The trajectory moves from standard generative baselines toward **Ref-RGAF** (Reference-Gated Affine Fusion), a candidate architecture that adds a deterministic control-reference path to the decoder.

**What this case study illustrates.** The trace records how the system evaluated modulation-based candidates, observed validation regressions in several VAE-style variants, and retained a reference-aware revision under the fixed validation protocol.

**Trajectory evidence.** The record links observed validation outcomes to subsequent local revisions and the candidate retained by the selection rule.

#### Closed-Loop Revision Trajectory

**Auditable closed-loop design record:  
candidate revision under validation feedback**



**(A) From Regression to Reference-Aware Generation** Generative candidates in the BBBC047 trajectory showed both validation gains and regressions. The run recorded possible underuse of the supplied control-reference information as a working diagnostic for subsequent candidate generation.

### Key Observation

**Observed pivot.** The retained revision injects a deterministic encoding  $h$  of the supplied control-reference profile alongside the stochastic latent variable in the decoder. The observed validation result supports retaining this reference-aware design.

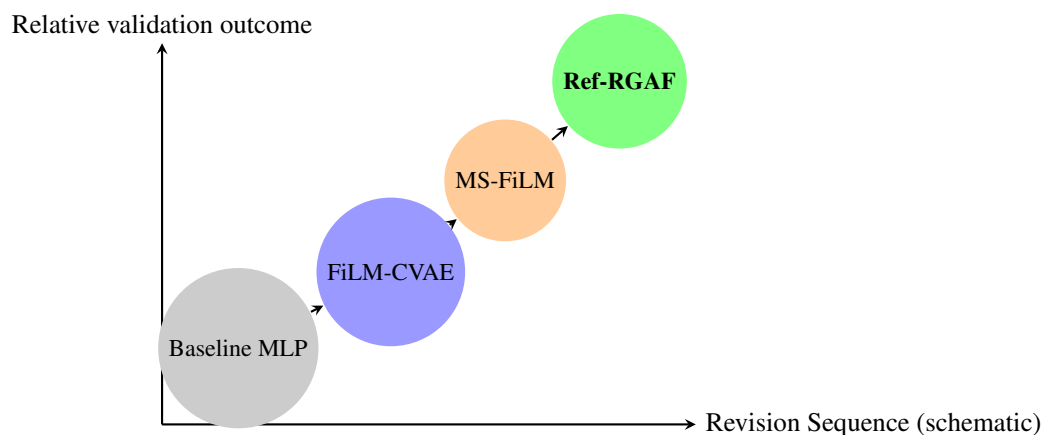


Figure 7: Auditable model-revision trajectory: the system moves from global modulation to multi-scale modulation and then to a reference-aware decoder.

**(B) Auditable Model-Revision Trajectory** We report the complete trajectory, including Phase 2 instability and Phase 3 revisions, to make validation-based candidate retention inspectable.

**Phase 1: Diagnosis and Constraint Identification.** The system identified data-quality issues and recorded preprocessing checks and split alignment before candidate evaluation (Table 10).

Table 10: **Phase 1 Execution Trace: Diagnosis and Constraint Identification.**

| Step | Activity            | Artifact     | Design / Operational Insight                                                                                                                              |
|------|---------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Data Ingestion      | HDF5/Pandas  | <b>Operational repair:</b> Fixed a <code>TypeError</code> caused by mixed numeric and string columns in the raw CSV.                                      |
| 2    | Numerical Diagnosis | Feature Scan | <b>Constraint definition:</b> Identified invalid-value warnings and added strict NaN checks and feature filtering to prevent training failures.           |
| 3    | Split Alignment     | Grouped CV   | <b>Protocol choice:</b> Grouped samples by exact SMILES identity, matching the SMILES-based split used for the reported controller and trajectory scores. |

**Phase 2: Generate and Execute.** Phase 2 recorded frequent code-generation and training failures. The baseline MLP had the highest validation PCC among eligible candidates and provided the reference for subsequent revisions (Table 11). This Phase 2 value is an exploratory setup-trace fit; the Phase 3 aggregate gains use a separate Phase 3 reference fit.

Table 11: Phase 2 Execution Trace: Establishing the Baseline.

| Iter | Strategy                        | Outcome               | PCC           | Design Analysis                                                                                                           |
|------|---------------------------------|-----------------------|---------------|---------------------------------------------------------------------------------------------------------------------------|
| 1    | Init                            | Crash                 | N/A           | Syntax error in setup node.                                                                                               |
| 2    | <b>Baseline MLP<sup>a</sup></b> | <b>Retained</b>       | <b>0.4118</b> | <b>Anchor candidate:</b> A standard MLP using median-normalized features provided a stable reference point.               |
| 3    | Innovation Var.                 | Regression            | 0.3834        | <b>Negative result:</b> A more complex candidate obtained a lower validation PCC and was rejected by the validation rule. |
| 4    | Training Loop                   | Repaired / Regression | 0.3707        | <b>Instability:</b> Training-node failure was repaired, but the candidate degraded.                                       |
| 5    | Final Attempt                   | Repaired / Regression | 0.3468        | <b>Regression:</b> Severe instability led the system to preserve Iteration 2 as the safe anchor.                          |

<sup>a</sup>Exploratory Phase 2 baseline; the Phase 3 trajectory uses the separately refit reference of 0.4120 shown in Table 13.

**Phase 3: Review and Revise.** Phase 3 evaluated modulation, gating, and reference-aware variants (Table 12). The retained Ref-RGAF candidate adds a deterministic control-reference path to the generative architecture.

Table 12: **Phase 3 Full Iteration Trace: Revision Logic.** “Improved” denotes an increase from the immediately preceding candidate; “Retained” denotes the best validation-selected candidate in this trace.

| Iter | Strategy         | Outcome         | PCC           | Design Analysis                                                                                                                                                                          |
|------|------------------|-----------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>FiLM-CVAE</b> | Improved        | 0.4148        | <b>Pivot:</b> Replaced a rigid ODE-style candidate with FiLM modulation, where drug embedding modulates the latent state.                                                                |
| 2    | Res-FiLM         | Regression      | 0.3416        | <b>Negative result:</b> The residual VAE obtained a lower validation PCC; the trace used possible underuse of control-reference information as a working diagnostic for later revisions. |
| 3    | Dose-Gated       | Improved        | 0.4095        | Candidate assumption: simple dose gating may be sufficient. Result: the candidate obtained a lower validation PCC than the retained anchor.                                              |
| 4    | TransMod         | Improved        | 0.4097        | The transformer-style fusion candidate obtained a lower validation PCC than the retained anchor.                                                                                         |
| 5    | <b>MS-FiLM</b>   | Improved        | <b>0.4237</b> | <b>Design cue:</b> Applying modulation at multiple decoder layers improved over bottleneck-only modulation.                                                                              |
| 6    | Gated-Attn       | Improved        | 0.4265        | Multiplicative gating improved validation PCC relative to the preceding candidates in this trace.                                                                                        |
| 7    | RGAF-FiLM        | Improved        | 0.4271        | Added affine fusion and a residual hypernetwork.                                                                                                                                         |
| 8    | <b>Ref-RGAF</b>  | <b>Retained</b> | <b>0.4368</b> | <b>Key revision:</b> Concatenated deterministic control-reference encoding $h$ with latent $z$ ; this candidate obtained the highest validation PCC in the trace.                        |
| 9    | State-Adaptive   | Regression      | 0.4249        | The state-adaptive fusion candidate obtained a lower validation PCC than Ref-RGAF.                                                                                                       |
| 10   | LoRA-FiLM        | Regression      | 0.4163        | The LoRA-style latent-rotation candidate obtained a lower validation PCC than Ref-RGAF.                                                                                                  |

**Failure Mode / Design Risk**

**Working diagnostic.** The residual VAE candidate used  $y = x + \text{Dec}(z)$  and obtained a lower validation PCC. The run record proposed possible underuse of the supplied control-reference information as a working diagnostic and used it to motivate reference injection.

**(C) Validation-Selected Design: Reference-Gated Affine Fusion** The retained **Ref-RGAF** candidate combines three implementation choices, described below as design rationales rather than independently established mechanisms.

**(1) Control-reference conditioning.** Ref-RGAF adds a deterministic reference path:

$$\text{Input to Decoder} = [z_{\text{stochastic}}, h_{\text{deterministic}}]. \quad (6)$$

This allows the decoder to condition its prediction on the supplied plate-matched DMSO-derived control-reference profile.

**(2) Affine dose modulation.** The drug effect is represented as a dose-conditioned affine transformation:

$$\text{DrugFeat} = (E_{\text{chem}} \odot \sigma(\text{Scale}(d))) + \text{Shift}(d). \quad (7)$$

This gives the model separate scale and shift pathways for response modulation.

**(3) Multi-scale modulation.** Drug modulation is applied at multiple decoder layers, allowing perturbation effects to enter at several representation levels.

**Design Rationale Encoded as Architecture**

**Design rationale.** The retained architecture reflects a task-specific modeling preference for preserving deterministic control-reference information while combining it with compound and dose features.

**(D) Validation Performance and Response-Sensitive Metrics** The retained candidate achieved a five-fold validation aggregate PCC of 0.4368 (Table 13). This is the validation aggregate reported for the selected CS-model in Table 1, not a separate final held-out evaluation. This outcome supports selecting the reference-aware revision within the reported trajectory and benchmark protocol.

Table 13: Trajectory of Candidate Performance (Phase 3).

| Iter             | Architecture Strategy | PCC           | $\Delta$ vs Base | RF-RMSE $\downarrow$ | Status          |
|------------------|-----------------------|---------------|------------------|----------------------|-----------------|
| Ref <sup>a</sup> | Baseline MLP          | 0.4120        | 0.00%            | 5.13                 | Reference       |
| 1                | FiLM-CVAE             | 0.4148        | +0.68%           | 5.07                 | Early gain      |
| 5                | MS-FiLM               | 0.4237        | +2.84%           | 5.01                 | Improved        |
| <b>8</b>         | <b>Ref-RGAF</b>       | <b>0.4368</b> | <b>+6.02%</b>    | <b>4.96</b>          | <b>Retained</b> |

<sup>a</sup>Phase 3 reference fit, separately refit from the exploratory Phase 2 baseline (0.4118); all Phase 3 gains in this table use 0.4120.

**Key Observation**

**Interpreting RF-RMSE.** Response-feature RMSE measures prediction error on the most perturbation-responsive morphology features, complementing global PCC when evaluating the retained candidate.

**(E) Summary** On BBBC047, CellScientist used a working diagnostic about control-reference use to propose the Ref-RGAF design. The trajectory preserves valid but underperforming candidates and links the reference-aware revision to execution results and validation-based selection.

Table 14: **Source-level statistics of the processed Cell Painting inputs used in our experiments.** Raw profiles include vehicle controls. Treated profiles are non-DMSO observations retained for response prediction. Exact non-control SMILES are counted within each benchmark, and raw Cell Painting feature counts precede split-specific feature filtering.

| Dataset | Cell line | Raw profiles | DMSO controls | Treated profiles | Exact non-control SMILES | Plates (all / treated) | Raw CP features |
|---------|-----------|--------------|---------------|------------------|--------------------------|------------------------|-----------------|
| BBBC036 | U2OS      | 18,540       | 3,528         | 15,012           | 1,916                    | 55 / 47                | 602             |
| BBBC047 | U2OS      | 59,275       | 16,302        | 42,973           | 11,415                   | 241 / 229              | 894             |
| CPG0016 | U2OS      | 235,307      | 32,837        | 202,470          | 82,282                   | 407 / 392              | 737             |

## C EXPERIMENTAL DETAILS

This appendix details the benchmark tasks, datasets, baselines, metrics, infrastructure, and configuration settings summarized in Section 4.1. It also specifies the evaluation protocols used to obtain the main-text results.

### C.1 BENCHMARK TASKS AND DATASETS

#### CELL PAINTING MORPHOLOGY PREDICTION

We evaluate the CellScientist workflow on three public Cell Painting datasets: BBBC036, BBBC047, and CPG0016. Prediction targets are CellProfiler features (Stirling et al., 2021) that quantify cellular and sub-cellular properties, including size, shape, texture, and fluorescence intensity. Feature-space readouts provide a common representation for perturbation-response modeling and reduce dependence on pixel-level image-processing choices. Each task includes an assay-context reference derived from vehicle controls according to its preprocessing protocol. This *control-reference profile* represents assay context; it is not a longitudinal pre-treatment measurement of the treated well.

BBBC036 and BBBC047 derive from the Cell Painting small-molecule collection of Bray et al. (Bray et al., 2017). CPG0016 is part of the JUMP Cell Painting resource (Chandrasekaran et al., 2023; Arevalo et al., 2024). All three use U2OS cells and span progressively larger sets of chemical perturbations. Table 14 reports source-profile counts for the processed inputs, distinguishing raw profiles from treated observations used for response prediction. Raw profiles include vehicle controls; treated profiles contain only retained non-DMSO observations. Exact non-control SMILES are counted within each benchmark, and raw feature counts precede split-specific filtering.

The morphology-response benchmarks use two group-based holdout settings, each assigning complete groups to a common fold. The SMILES-based setting groups records by exact recorded SMILES, whereas the plate-based setting groups records by `Metadata_Plate`. These settings evaluate exact-structure and experimental-plate holdout, respectively; neither is a scaffold split.

#### LINCS2020 TRANSCRIPTOMIC PERTURBATION PREDICTION

For transcriptomic perturbation prediction, we use gene-expression profiles from the next-generation Connectivity Map (CMap) project, which measures large-scale transcriptional responses with the L1000 platform (Subramanian et al., 2017). Raw and processed profiles are obtained through the LINCS data portal (Stathias et al., 2020), following recommended access and preprocessing protocols (Xie et al., 2022). We construct two benchmark variants to evaluate different generalization regimes: a controlled **Seven Cell Lines** subset and a large-scale **Full Data** benchmark.

The **Seven Cell Lines** subset contains 355 compounds profiled across seven frequently assayed cell lines: A375, HELA, PC3, MCF7, HT29, YAPC, and HA1E. This dense subspace provides high compound-cell coverage and enables a chemical-blind split, where scored compounds are unseen in the corresponding training folds. It therefore measures generalization to held-out compound identities. The **Full Data** benchmark contains 8,316 compounds across 164 cell lines, yielding 78,569 compound-cell expression profiles. This setting is substantially sparser and more heterogeneous and measures performance under long-tail cell-line coverage and diverse perturbation contexts. Dataset statistics are summarized in Table 15.

Table 15: Statistics of LINCS L1000 benchmarks.

| Dataset          | # Compounds | # Cell Lines | # Samples | Characteristics       |
|------------------|-------------|--------------|-----------|-----------------------|
| Seven Cell Lines | 355         | 7            | ~2,485    | Dense; chemical-blind |
| Full Data        | 8,316       | 164          | 78,569    | Sparse; long-tail     |

Table 16: Statistics of single-cell benchmarks used in our evaluations. Readout dimensions are reported where the evaluation uses a fixed feature matrix; Papalexi RNA and protein are evaluated as separate readouts.

| Benchmark | Assay / task           | Cells  | Readout and condition statistics                                                                              |
|-----------|------------------------|--------|---------------------------------------------------------------------------------------------------------------|
| Norman    | CRISPRa scRNA-seq      | 91,205 | 5,045 RNA genes; 284 conditions (152 single-gene, 131 two-gene, and 1 control)                                |
| Papalexi  | CRISPR-Cas9 ECCITE-seq | 20,729 | RNA and surface-protein ADT readouts; 25 targeted genes plus non-targeting control and 98 perturbation labels |

## SINGLE-CELL PERTURBATION PREDICTION

We further evaluate the CellScientist workflow on public single-cell benchmarks covering genetic perturbation and multimodal perturbation responses. These datasets complement morphology and L1000 evaluations with measurements at single-cell resolution and response-sensitive prediction targets.

**Norman** (Norman et al., 2019) is a pooled CRISPRa scRNA-seq dataset designed to study the effects of single and combinatorial gene activation. It is commonly used to evaluate transcriptional-response prediction for genetic interventions, including combinations. **Papalexi** (Papalexi et al., 2021) is a CRISPR-Cas9 ECCITE-seq dataset that jointly profiles RNA and surface-protein responses after genetic perturbation, enabling evaluation on both transcriptomic and antibody-derived-tag readouts. Table 16 summarizes the processed benchmark statistics used in the corresponding evaluations.

We report task-specific workflow results on Norman, Papalexi RNA, and Papalexi Protein in Appendix F.3. Each readout is evaluated under its dataset-specific protocol.

## C.2 BASELINES

## MORPHOLOGY BASELINES

For morphology prediction, we organize baselines into three comparison groups used consistently in the result tables.

**Existing reference methods** include strong feature-space predictors:

- **TabR** (Gorishniy et al., 2024): A tabular prediction model with a retrieval component that uses nearest-neighbor information at inference time.
- **RealMLP** (Holzmüller et al., 2024): A carefully tuned MLP baseline designed to provide strong and stable performance on tabular benchmarks.
- **RF-TD** (Holzmüller et al., 2024): A Random Forest baseline with tuned default settings for competitive tabular prediction.

**Search-based baselines** provide generic model-selection reference points under the corresponding benchmark split, target, and metric definitions:

- **FLAML** (Wang et al., 2021): An AutoML framework that searches over model and hyperparameter configurations under a configured budget.
- **Random Search** (Gijssbers et al., 2024): A generic search baseline that samples candidate configurations independently.

**Agent-designed baselines** test whether an existing automated modeling workflow is sufficient:

- **CellForge** (Tang et al., 2025): A multi-agent automated modeling framework for proposing, training, and selecting predictive models.

These groups compare fixed supervised predictors, generic model-selection procedures, and the CellForge automated perturbation-modeling workflow.

#### TRANSCRIPTOMIC AND SINGLE-CELL BASELINES

For LINCS2020 transcriptomic perturbation prediction, we compare against representative domain-specific baselines:

- **DLEPS** (Zhu et al., 2021): A CNN-based model that maps chemical structures to gene-expression changes with cell-line embeddings.
- **DeepCE** (Pham et al., 2021): A graph neural network framework that combines chemical substructure representations with cell-line features.
- **CIGER** (Pham et al., 2022): A collaborative filtering approach that models drug-cell-dose interactions with graph convolutional networks.
- **MultiDCP** (Wu et al., 2022): A multi-view deep learning model with separate drug and cell-line encoders and contrastive fusion.
- **TranSiGen** (Tong et al., 2024): A VAE-based generative model that learns latent representations for denoising and reconstructing perturbation profiles.

For the workflow-instantiation results on Norman and Papalex, we compare against **CellForge** (Tang et al., 2025) under the dataset-specific evaluation procedures.

### C.3 EVALUATION METRICS

#### MORPHOLOGY METRICS

For morphology prediction, we evaluate both global profile fidelity and perturbation-sensitive feature recovery. Given the stacked prediction and target matrices  $\hat{\mathbf{Y}}$  and  $\mathbf{Y}$  for an evaluation partition, global PCC is  $\text{corr}(\text{vec}(\hat{\mathbf{Y}}), \text{vec}(\mathbf{Y}))$ ; MSE and  $R^2$  are computed over the corresponding entries. This same global-PCC functional is used in the high-freedom morphology benchmark and the controlled audit, while the fitted predictor, partition aggregation, and experimental replication follow their respective protocols. MSE captures numerical reconstruction error, PCC measures global profile association, and  $R^2$  evaluates explained feature variance; we report all three because high-dimensional morphology profiles can differ in scale, calibration, and variance across datasets.

To further assess whether a model captures the most responsive morphology changes, we compute response-feature metrics on the top- $K$  perturbed features. For each sample  $i$ , let  $\mathbf{y}_{i,\text{gt}}$  be the ground-truth response vector and  $\boldsymbol{\mu}_{\text{ctrl}}$  the mean control profile. We select responsive features using only ground-truth deviation from control:

$$\mathcal{F}_i^{(K)} = \text{topk}_j \left( \left| \mathbf{y}_{i,\text{gt}}^{(j)} - \boldsymbol{\mu}_{\text{ctrl}}^{(j)} \right| \right), \quad K \in \{20, 50\}.$$

We then compute  $\text{RMSE}_{\text{RF-}K}$  and  $\text{PCC}_{\text{RF-}K}$  within  $\mathcal{F}_i^{(K)}$ , where RF denotes response feature. The feature subset is determined solely from ground truth and control profiles, independently of model outputs. These metrics provide complementary evaluation views, while PCC remains the configured model-selection metric.

#### TRANSCRIPTOMIC AND SINGLE-CELL METRICS

For LINCS2020 transcriptomic prediction, RMSE and PCC measure global expression-profile fidelity. Positive and Negative P@100 measure recovery of the most strongly upregulated and downregulated genes, respectively. These metrics complement whole-profile accuracy with the recovery of strongly responding genes.

For single-cell perturbation prediction, we report PCC and  $\text{PCC}_{\text{DE}}$ . PCC measures global agreement between predicted and observed response profiles, while  $\text{PCC}_{\text{DE}}$  restricts the evaluation to differentially

expressed genes and therefore emphasizes perturbation-sensitive recovery. Together, these metrics separate overall profile reconstruction from recovery of key response signals.

#### C.4 IMPLEMENTATION DETAILS

##### IMPLEMENTATION AND EXECUTION CONFIGURATION

Table 17 reports the environments used for the high-freedom workflow and the controlled audit. Methods compared within each experiment use the same hardware and software environment. Table 18 separates open-workflow settings from those used to compare controllers within the fixed candidate language.

Table 17: **Computational infrastructure and software environment.**

| Component                      | Specification                              |
|--------------------------------|--------------------------------------------|
| High-freedom workflow GPU      | NVIDIA RTX 5880 Ada Generation (48GB VRAM) |
| High-freedom workflow software | Python 3.11.14; PyTorch 2.0.1+cu118        |
| Controlled-audit GPU           | NVIDIA H100 (80GB HBM3)                    |
| Controlled-audit software      | Python 3.10.20; PyTorch 2.3.1+cu121        |
| Operating system               | Linux                                      |

##### PROTOCOL CONTROLS AND ROUTING DETAILS

This section specifies evaluation semantics and feedback routing for the high-freedom workflow. Within each benchmark, methods use the reported split, target, and metric definitions. Fixed predictors follow their model protocols, search baselines use configured search procedures, and CellForge follows its agent workflow. CellScientist uses 10 revision steps with up to 3 local implementation repairs per step. Candidates are eligible for selection only after passing the configured checks and completing scoring; rejected and failed candidates remain in the run history.

PDR associates structured execution and validation diagnostics with proposed edits to task components. Diagnostic records identify one or more edit targets; LCA then checks the resulting edits against the configured task contract. Table 19 summarizes representative mappings. Listing 1 gives a normalized illustration reconstructed from the BBBC047 trajectory.

Table 18: CellScientist controller and execution settings.

| Configuration                       | Value                               | Role in the workflow                                                |
|-------------------------------------|-------------------------------------|---------------------------------------------------------------------|
| Open-workflow backbone              | Gemini 3 Pro                        | Default backbone for high-freedom model exploration                 |
| Open-workflow context length        | 40,000 tokens                       | Maintained interaction context limit                                |
| Open-workflow temperatures          | 0.5 analysis; 0.7 proposal          | Sampling settings for analysis and generative revision              |
| Open-workflow prompt views          | 4                                   | Robust scaling, PCA, feature-network, and enrichment views          |
| Open-workflow iteration budget      | 10 revisions                        | Maximum number of high-freedom revision steps                       |
| Open-workflow repair budget         | 3 rounds                            | Configured debugging budget for runtime or interface errors         |
| Controlled-audit backbones          | Gemini 3 Pro; Qwen2.5-0.5B-Instruct | Manuscript-default and open-weight reproduction                     |
| Controlled-audit decoding           | Temperature 0                       | Fixed within every matched controller comparison                    |
| Controlled-audit candidate language | 768 executable candidates           | Shared predictor configurations available to every policy           |
| Controlled-audit fitting            | Seeded randomized low-rank fitting  | Identical fitting seed for every policy within each paired run      |
| Controlled-audit budgets            | $B \in \{1, 3, 5, 10\}$             | Trained and evaluated candidates, including $h_0$                   |
| Validation selection metric         | Global PCC                          | Metric used by $\text{Select}_{\text{val}}$ in the controlled audit |
| Controlled-audit retention view     | Mean feedback/selection PCC         | Identical candidate-retention rule for every matched policy         |

Table 19: **Representative manuscript-level PDR mappings.** PDR maps diagnostic signals to proposed local edit targets; LCA checks the resulting edits against protected evaluation semantics.

| Diagnostic                              | Editable region                                      | Protected semantics                                |
|-----------------------------------------|------------------------------------------------------|----------------------------------------------------|
| Interface failure                       | Data adapter or implementation interface             | Split, target, and metric definitions              |
| Metric-reporting failure                | Evaluation adapter                                   | Metric formulas and fold protocol                  |
| Validation collapse                     | Fusion, decoder, objective, or training node         | Dataset split, target definition, and output space |
| Weak perturbation sensitivity           | Perturbation encoding, dose encoding, or fusion edge | Evaluation code and response-feature definition    |
| Suspected underuse of control reference | Encoder state, decoder input, or fusion component    | Preprocessing, split, target, and metrics          |
| Batch-shortcut risk                     | Representation or training-regularization component  | Configured split and evaluation rules              |

```
{
  "dataset": "BBBC047",
  "iteration": 2,
  "diagnostic": {
    "type": "suspected_control_reference_underuse",
    "candidate": "Res-FiLM",
    "execution_status": "valid",
    "rule_status": "passed",
    "selection_metric": "PCC",
    "candidate_score": 0.3416,
    "best_previous_score": 0.4148,
    "verdict": "valid_but_regressed",
    "summary": "Validation PCC dropped after a residual VAE-style update; control-reference use was recorded as a working diagnostic."
  },
}
```

```

"routed_address": {
  "address_type": "proposed_edit_targets",
  "editable_components": [
    "model_architecture",
    "encoder_state_representation",
    "decoder_input",
    "fusion_module"
  ],
  "routing_reason": "valid execution with degraded validation performance
    and a working diagnostic about control-reference use"
},
"allowed_edit": {
  "operation": "test_reference_aware_decoder_path",
  "protected_components": [
    "dataset_split",
    "target_definition",
    "feature_preprocessing",
    "validation_protocol",
    "evaluation_metrics",
    "fold_protocol"
  ]
},
"revised_candidate": {
  "name": "Ref-RGAF",
  "design_change": "concatenate deterministic control-reference encoding
    h with stochastic latent code z before decoding",
  "intended_effect": "retain supplied control-reference information while
    allowing compound-dose modulation"
},
"lca_check": [
  "interface_check",
  "shape_check",
  "leakage_check",
  "metric_completeness"
]
}

```

Listing 1: Normalized manuscript-level illustration reconstructed from the BBBC047 trajectory.

## LLM BACKBONES

The matched-budget audit uses Gemini 3 Pro for the manuscript-default runs and repeats the controller comparison with the open-weight Qwen2.5-0.5B-Instruct model. Within each backbone, every controller receives the same decoding configuration and candidate budget. Table 20 records the public model identities used in this audit.

**Controller-call accounting.** Gemini completed all 720 logical controller decisions with valid allow-listed responses and no fallback. Three transient transport attempts were retried successfully and remain included in request accounting. Qwen completed all 720 local HTTP requests successfully. Its schema validator invoked the registered legal fallback for 10/180 CellScientist requests and 70/180 AIDE-style requests after non-allow-listed field pairings. Flat and Random routing required no fallback. All such events remain in the reported trajectories and candidate budgets.

Table 20: **LLM backbones used in the controlled framework audit.** Gemini 3 Pro is the manuscript-default backbone; Qwen2.5-0.5B-Instruct provides an open-weight reproduction.

| Main-text Label | Full Backbone Identity | Provider / Family | Source Link                                                                                                       |
|-----------------|------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------|
| Gemini 3 Pro    | Gemini 3 Pro           | Google Gemini     | <a href="https://blog.google/products/gemini/gemini-3">https://blog.google/products/gemini/gemini-3</a>           |
| Qwen 0.5B       | Qwen2.5-0.5B-Instruct  | Qwen              | <a href="https://huggingface.co/Qwen/Qwen2.5-0.5B-Instruct">https://huggingface.co/Qwen/Qwen2.5-0.5B-Instruct</a> |

## D CONTROLLER-BACKBONE REPRODUCTION

Table 21: **Task-level reproduction of structured routing across controller backbones.** The upper panel reports mean $\pm$ sample standard deviation in held-out global PCC over the same five paired controller and fitting seeds. The lower panel reports CellScientist-minus-Random effects with paired-setting 95%  $t$ -confidence intervals across the four task/split settings, computed from unrounded run-level values; confidence intervals are shown in smaller type. Gemini 3 Pro and the open-weight Qwen2.5-0.5B-Instruct use the same settings, candidate language, fitting procedure, and evaluation budgets. Bold marks the better mean within each backbone, setting, and budget.

| Backbone                                                              | Dataset                | Split  | $B = 3$                   |                            | $B = 5$                   |                            | $B = 10$                   |                            |
|-----------------------------------------------------------------------|------------------------|--------|---------------------------|----------------------------|---------------------------|----------------------------|----------------------------|----------------------------|
|                                                                       |                        |        | Random                    | CellScientist              | Random                    | CellScientist              | Random                     | CellScientist              |
| Gemini 3 Pro                                                          | BBBC036                | Plate  | 0.2250 $\pm$ 0.0493       | <b>0.2650</b> $\pm$ 0.0005 | 0.2277 $\pm$ 0.0502       | <b>0.2674</b> $\pm$ 0.0003 | 0.2626 $\pm$ 0.0077        | <b>0.2693</b> $\pm$ 0.0012 |
|                                                                       | BBBC036                | SMILES | 0.2052 $\pm$ 0.0413       | <b>0.2336</b> $\pm$ 0.0016 | 0.2049 $\pm$ 0.0417       | <b>0.2346</b> $\pm$ 0.0009 | <b>0.2347</b> $\pm$ 0.0016 | 0.2346 $\pm$ 0.0009        |
|                                                                       | BBBC047                | Plate  | 0.3032 $\pm$ 0.0500       | <b>0.3396</b> $\pm$ 0.0004 | 0.3066 $\pm$ 0.0514       | <b>0.3457</b> $\pm$ 0.0004 | 0.3458 $\pm$ 0.0027        | <b>0.3481</b> $\pm$ 0.0009 |
|                                                                       | BBBC047                | SMILES | 0.3131 $\pm$ 0.0536       | <b>0.3529</b> $\pm$ 0.0010 | 0.3207 $\pm$ 0.0582       | <b>0.3614</b> $\pm$ 0.0003 | 0.3668 $\pm$ 0.0073        | <b>0.3751</b> $\pm$ 0.0017 |
| Qwen2.5-0.5B                                                          | BBBC036                | Plate  | 0.2250 $\pm$ 0.0493       | <b>0.2566</b> $\pm$ 0.0024 | 0.2277 $\pm$ 0.0502       | <b>0.2566</b> $\pm$ 0.0024 | 0.2626 $\pm$ 0.0077        | <b>0.2657</b> $\pm$ 0.0035 |
|                                                                       | BBBC036                | SMILES | 0.2052 $\pm$ 0.0413       | <b>0.2357</b> $\pm$ 0.0004 | 0.2049 $\pm$ 0.0417       | <b>0.2357</b> $\pm$ 0.0004 | <b>0.2347</b> $\pm$ 0.0016 | 0.2346 $\pm$ 0.0009        |
|                                                                       | BBBC047                | Plate  | 0.3032 $\pm$ 0.0500       | <b>0.3429</b> $\pm$ 0.0003 | 0.3066 $\pm$ 0.0514       | <b>0.3425</b> $\pm$ 0.0009 | <b>0.3458</b> $\pm$ 0.0027 | 0.3438 $\pm$ 0.0018        |
|                                                                       | BBBC047                | SMILES | 0.3131 $\pm$ 0.0536       | <b>0.3570</b> $\pm$ 0.0005 | 0.3207 $\pm$ 0.0582       | <b>0.3615</b> $\pm$ 0.0005 | 0.3668 $\pm$ 0.0073        | <b>0.3707</b> $\pm$ 0.0002 |
| <i>CellScientist – Random: mean effect with paired-setting 95% CI</i> |                        |        |                           |                            |                           |                            |                            |                            |
| Backbone                                                              | Contrast               |        | $B = 3$                   |                            | $B = 5$                   |                            | $B = 10$                   |                            |
| Gemini 3 Pro                                                          | CellScientist – Random |        | +0.0361 [+0.0275,+0.0448] |                            | +0.0373 [+0.0292,+0.0454] |                            | +0.0043 [-0.0018,+0.0104]  |                            |
| Qwen2.5-0.5B-Instruct                                                 | CellScientist – Random |        | +0.0364 [+0.0261,+0.0467] |                            | +0.0341 [+0.0256,+0.0426] |                            | +0.0012 [-0.0032,+0.0056]  |                            |

## E DETAILED LCA AND PDR AUDIT RESULTS

Table 4 summarizes the HRT, LCA, and PDR results in the main text. Here we report LCA outcomes and PDR results by component address, including registered case counts, decisions, localization, repair success, unrelated edits, and repair rounds.

### E.1 REGISTERED LCA OUTCOME AUDIT

The LCA audit includes 40 registered violation cases and 20 legal controls. Each positive case contains one prespecified violation, and each legal control contains none. LCA repaired 15 pre-run cases, blocked 15 pre-run violations, and rejected 10 post-run violations. Legal controls remained unchanged, yielding 60/60 correct decisions and 40/40 contained violations. Thirty invalid evaluations were avoided before execution.

### E.2 HELD-OUT REGISTERED-FAULT PDR AUDIT

The PDR audit uses five calibration cases and 15 held-out faults, with the split fixed before evaluation. Each policy is evaluated on the same faults, with three cases per component address. Flat retry and Random routing provide comparison policies for localization and repair. Structured PDR achieved 100% Top-1 and Top-2 localization at every address. It completed every repair with 0 unrelated edits and 1.00 mean repair round.

Table 22: **Registered LCA audit grouped by outcome.** Each positive case contains one prespecified violation and each legal control contains none. The table separates violations that can be restored to an eligible candidate from violations that must be safely rejected before or after execution. “Eligible” denotes eligibility after LCA, and “Avoided” counts invalid evaluations avoided. Across the 40 positive and 20 negative cases, violation recall, exact-code accuracy, legal-case specificity, and repair-or-safe-block containment are all 100%.

| LCA outcome    | Registered cases                                                                  | Stage | Cases     | Detected / accepted   | Observed action        | Eligible              | Avoided   |
|----------------|-----------------------------------------------------------------------------------|-------|-----------|-----------------------|------------------------|-----------------------|-----------|
| Repair         | Protected-state mutation; interface mismatch; route mismatch                      | Pre   | 15        | <b>15/15 detected</b> | <b>15/15 repaired</b>  | <b>15/15</b>          | 15        |
| Pre-run block  | Out-of-domain candidate; non-atomic edit; repair budget exhausted                 | Pre   | 15        | <b>15/15 detected</b> | <b>15/15 blocked</b>   | 0/15                  | 15        |
| Post-run block | Output-shape mismatch; non-finite output                                          | Post  | 10        | <b>10/10 detected</b> | <b>10/10 blocked</b>   | 0/10                  | 0         |
| Legal controls | Contract-preserving candidates across registered interfaces and component domains | Pre   | 20        | <b>20/20 accepted</b> | Unchanged              | <b>20/20</b>          | 0         |
| <b>Overall</b> | All registered violation and legal-control cases                                  |       | <b>60</b> | <b>60/60 correct</b>  | <b>40/40 contained</b> | <b>35/60 eligible</b> | <b>30</b> |

Table 23: **Held-out registered-fault PDR audit by component address.** The fault-to-address split is frozen before evaluation: five cases are used for calibration and the remaining 15 for reporting. Each held-out address has three registered cases. Structured PDR uses Gemini 3 Pro to select an allow-listed address from the diagnostic shortlist. Unrepaired cases are censored at the five-round repair budget. Best values within each address are **bolded**; ties share emphasis.

| Routing policy | Expected address          | Cases | Top-1 accuracy $\uparrow$ | Top-2 recall $\uparrow$ | Repair success $\uparrow$ | Mean unrelated edits $\downarrow$ | Mean repair rounds $\downarrow$ |
|----------------|---------------------------|-------|---------------------------|-------------------------|---------------------------|-----------------------------------|---------------------------------|
| Flat retry     | Conditioning              | 3     | 66.7%                     | N/A                     | 0.0%                      | 4.33                              | 5.00                            |
|                | Response parameterization | 3     | 0.0%                      | N/A                     | <b>100%</b>               | 1.00                              | 2.00                            |
|                | Representation            | 3     | 0.0%                      | N/A                     | 0.0%                      | 4.00                              | 5.00                            |
|                | Decoder                   | 3     | 0.0%                      | N/A                     | 0.0%                      | 4.33                              | 5.00                            |
|                | Regularization            | 3     | 0.0%                      | N/A                     | 66.7%                     | 2.33                              | 3.67                            |
|                | All held-out              | 15    | 13.3%                     | N/A                     | 33.3%                     | 3.20                              | 4.13                            |
| Random routing | Conditioning              | 3     | 0.0%                      | 0.0%                    | 0.0%                      | 5.00                              | 5.00                            |
|                | Response parameterization | 3     | 33.3%                     | 33.3%                   | 66.7%                     | 2.67                              | 3.33                            |
|                | Representation            | 3     | 0.0%                      | 0.0%                    | 66.7%                     | 3.33                              | 4.00                            |
|                | Decoder                   | 3     | 0.0%                      | 33.3%                   | <b>100%</b>               | 2.33                              | 3.33                            |
|                | Regularization            | 3     | 0.0%                      | 0.0%                    | 66.7%                     | 3.00                              | 3.67                            |
|                | All held-out              | 15    | 6.7%                      | 13.3%                   | 60.0%                     | 3.27                              | 3.87                            |
| Structured PDR | Conditioning              | 3     | <b>100%</b>               | <b>100%</b>             | <b>100%</b>               | <b>0.00</b>                       | <b>1.00</b>                     |
|                | Response parameterization | 3     | <b>100%</b>               | <b>100%</b>             | <b>100%</b>               | <b>0.00</b>                       | <b>1.00</b>                     |
|                | Representation            | 3     | <b>100%</b>               | <b>100%</b>             | <b>100%</b>               | <b>0.00</b>                       | <b>1.00</b>                     |
|                | Decoder                   | 3     | <b>100%</b>               | <b>100%</b>             | <b>100%</b>               | <b>0.00</b>                       | <b>1.00</b>                     |
|                | Regularization            | 3     | <b>100%</b>               | <b>100%</b>             | <b>100%</b>               | <b>0.00</b>                       | <b>1.00</b>                     |
|                | All held-out              | 15    | <b>100%</b>               | <b>100%</b>             | <b>100%</b>               | <b>0.00</b>                       | <b>1.00</b>                     |

## F ADDITIONAL EXPERIMENTAL ANALYSIS

This appendix characterizes the high-freedom workflow, evaluates recovery of responsive morphology features, and reports additional single-cell results. These analyses complement the main results under their respective evaluation protocols.

### F.1 END-TO-END OPEN-WORKFLOW CHARACTERIZATION

**Open-workflow evidence.** Tables 1 and 2 report task-specific predictors selected by the high-freedom CellScientist workflow under morphology and transcriptomic protocols. Figures 4 and 5 show fold-level comparisons and non-monotonic revision trajectories, respectively. These results assess predictor quality; the main-text RQ2 audit separately compares controller policies within a fixed candidate language.

The RQ2 study compares controller policies within a fixed executable language. Table 24 complements that audit with an end-to-end comparison in which implementations and selected model families can vary. Cumulative configurations share the task specification, split, validation metric, iteration budget, repair budget, execution timeout, and environment. The comparison records execution success, rule-compliant success, predictive quality, and runtime as structured task state, configured checks, and history-conditioned revision are added.

Table 24: **End-to-end open-workflow characterization.** SR denotes execution success, R-SR denotes rule-compliant success, and Avg/Best PCC summarize validation performance over eligible runs. Time reports wall-clock runtime. This high-freedom characterization complements the fixed-language comparisons in RQ2.

| ID | Method Configuration                    | BBBC036 (SMILES-based Split) |                 |                    |                     |                       | BBBC047 (SMILES-based Split) |                 |                    |                     |                       |
|----|-----------------------------------------|------------------------------|-----------------|--------------------|---------------------|-----------------------|------------------------------|-----------------|--------------------|---------------------|-----------------------|
|    |                                         | SR $\uparrow$                | R-SR $\uparrow$ | Avg PCC $\uparrow$ | Best PCC $\uparrow$ | Time (h) $\downarrow$ | SR $\uparrow$                | R-SR $\uparrow$ | Avg PCC $\uparrow$ | Best PCC $\uparrow$ | Time (h) $\downarrow$ |
| M0 | Prompt + LLM + Code Executor            | 0.526                        | 0.526           | 0.1357             | 0.1682              | 10.25                 | 0.368                        | 0.368           | 0.1422             | 0.1752              | 15.58                 |
| M1 | M0 + Re-prompting (Naive)               | 0.263                        | 0.263           | 0.2156             | 0.2485              | 17.96                 | 0.210                        | 0.210           | 0.2267             | 0.2581              | 27.29                 |
| M2 | M1 + Structured Task State (HRT)        | 0.947                        | 0.684           | 0.2771             | 0.3050              | 10.34                 | 0.941                        | 0.588           | 0.2472             | 0.2775              | 15.96                 |
| M3 | M2 + Protected Editing / Checks (LCA)   | <b>1.000</b>                 | 0.737           | 0.3183             | 0.3214              | 8.46                  | <b>1.000</b>                 | 0.684           | 0.3315             | 0.3962              | 13.38                 |
| M4 | M3 + History-Conditioned Revision (PDR) | <b>1.000</b>                 | <b>0.789</b>    | <b>0.3413</b>      | <b>0.3688</b>       | <b>7.93</b>           | <b>1.000</b>                 | <b>0.714</b>    | <b>0.3539</b>      | <b>0.4368</b>       | <b>12.58</b>          |

Table 25: **Response-sensitive morphology metrics.** Response-feature metrics evaluate prediction quality on the top- $K$  most responsive morphology features. CS-model has the best mean values among the listed methods under both SMILES-based and plate-based splits in this table. Best mean values are **bolded**.

| Method                       | Venue      | SMILES-based Split            |                               |                               |                               | Plate-based Split             |                               |                               |                               |
|------------------------------|------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
|                              |            | RMSE <sub>RF-20</sub> ↓       | RMSE <sub>RF-50</sub> ↓       | PCC <sub>RF-20</sub> ↑        | PCC <sub>RF-50</sub> ↑        | RMSE <sub>RF-20</sub> ↓       | RMSE <sub>RF-50</sub> ↓       | PCC <sub>RF-20</sub> ↑        | PCC <sub>RF-50</sub> ↑        |
| BBBC036                      |            |                               |                               |                               |                               |                               |                               |                               |                               |
| # Existing reference methods |            |                               |                               |                               |                               |                               |                               |                               |                               |
| TabR                         | ICLR'24    | 4.1306 <sub>±0.03</sub>       | 3.3883 <sub>±0.04</sub>       | 0.2742 <sub>±0.01</sub>       | 0.2584 <sub>±0.01</sub>       | 3.8974 <sub>±0.12</sub>       | 3.1610 <sub>±0.07</sub>       | 0.3784 <sub>±0.05</sub>       | 0.3783 <sub>±0.05</sub>       |
| RealMLP                      | NeurIPS'24 | 3.9910 <sub>±0.03</sub>       | 3.2980 <sub>±0.03</sub>       | 0.3170 <sub>±0.01</sub>       | 0.2929 <sub>±0.01</sub>       | 3.5303 <sub>±0.13</sub>       | 2.8547 <sub>±0.08</sub>       | 0.4636 <sub>±0.03</sub>       | 0.4681 <sub>±0.01</sub>       |
| RF-TD                        | NeurIPS'24 | 4.1499 <sub>±0.02</sub>       | 3.4356 <sub>±0.03</sub>       | 0.2445 <sub>±0.01</sub>       | 0.2227 <sub>±0.01</sub>       | 3.9590 <sub>±0.13</sub>       | 3.2927 <sub>±0.07</sub>       | 0.5268 <sub>±0.05</sub>       | 0.4649 <sub>±0.05</sub>       |
| □ Search-based baselines     |            |                               |                               |                               |                               |                               |                               |                               |                               |
| FLAML                        | MLSys'21   | 3.9892 <sub>±0.04</sub>       | 3.2897 <sub>±0.04</sub>       | 0.3297 <sub>±0.01</sub>       | 0.3079 <sub>±0.01</sub>       | 3.3658 <sub>±0.11</sub>       | 2.7702 <sub>±0.08</sub>       | 0.6486 <sub>±0.03</sub>       | 0.6705 <sub>±0.03</sub>       |
| Random Search                | JMLR'24    | 3.9948 <sub>±0.05</sub>       | 3.2964 <sub>±0.05</sub>       | 0.3218 <sub>±0.01</sub>       | 0.2978 <sub>±0.01</sub>       | 3.4316 <sub>±0.12</sub>       | 2.8357 <sub>±0.08</sub>       | 0.6235 <sub>±0.04</sub>       | 0.6468 <sub>±0.04</sub>       |
| △ Agent-designed methods     |            |                               |                               |                               |                               |                               |                               |                               |                               |
| CellForge                    | arXiv'25   | 3.9903 <sub>±0.05</sub>       | 3.2928 <sub>±0.05</sub>       | 0.3262 <sub>±0.01</sub>       | 0.3034 <sub>±0.01</sub>       | 3.3869 <sub>±0.10</sub>       | 2.7883 <sub>±0.08</sub>       | 0.6574 <sub>±0.03</sub>       | 0.6813 <sub>±0.03</sub>       |
| CS-model                     | Ours       | <b>3.9865<sub>±0.09</sub></b> | <b>3.2849<sub>±0.08</sub></b> | <b>0.3339<sub>±0.01</sub></b> | <b>0.3139<sub>±0.01</sub></b> | <b>3.2817<sub>±0.11</sub></b> | <b>2.7046<sub>±0.09</sub></b> | <b>0.6856<sub>±0.02</sub></b> | <b>0.7082<sub>±0.02</sub></b> |
| BBBC047                      |            |                               |                               |                               |                               |                               |                               |                               |                               |
| # Existing reference methods |            |                               |                               |                               |                               |                               |                               |                               |                               |
| TabR                         | ICLR'24    | 6.2132 <sub>±0.02</sub>       | 4.6263 <sub>±0.01</sub>       | 0.3355 <sub>±0.01</sub>       | 0.3898 <sub>±0.01</sub>       | 6.2120 <sub>±0.05</sub>       | 4.6035 <sub>±0.04</sub>       | 0.3398 <sub>±0.01</sub>       | 0.4009 <sub>±0.01</sub>       |
| RealMLP                      | NeurIPS'24 | 5.8479 <sub>±0.02</sub>       | 4.4168 <sub>±0.01</sub>       | 0.3678 <sub>±0.01</sub>       | 0.4851 <sub>±0.01</sub>       | 6.1791 <sub>±0.06</sub>       | 4.5301 <sub>±0.04</sub>       | 0.3836 <sub>±0.02</sub>       | 0.4270 <sub>±0.01</sub>       |
| RF-TD                        | NeurIPS'24 | 6.0601 <sub>±0.03</sub>       | 4.5652 <sub>±0.02</sub>       | 0.3742 <sub>±0.01</sub>       | 0.4153 <sub>±0.01</sub>       | 6.0934 <sub>±0.05</sub>       | 4.5715 <sub>±0.04</sub>       | 0.3591 <sub>±0.02</sub>       | 0.4063 <sub>±0.01</sub>       |
| □ Search-based baselines     |            |                               |                               |                               |                               |                               |                               |                               |                               |
| FLAML                        | MLSys'21   | 5.2847 <sub>±0.04</sub>       | 4.0209 <sub>±0.03</sub>       | 0.3685 <sub>±0.01</sub>       | 0.4979 <sub>±0.01</sub>       | 5.5128 <sub>±0.05</sub>       | 4.0879 <sub>±0.04</sub>       | 0.3892 <sub>±0.02</sub>       | 0.5096 <sub>±0.01</sub>       |
| Random Search                | JMLR'24    | 5.4362 <sub>±0.05</sub>       | 4.1328 <sub>±0.04</sub>       | 0.3601 <sub>±0.01</sub>       | 0.4887 <sub>±0.01</sub>       | 5.6315 <sub>±0.06</sub>       | 4.1746 <sub>±0.04</sub>       | 0.3838 <sub>±0.02</sub>       | 0.4983 <sub>±0.01</sub>       |
| △ Agent-designed methods     |            |                               |                               |                               |                               |                               |                               |                               |                               |
| CellForge                    | arXiv'25   | 5.1164 <sub>±0.04</sub>       | 3.8946 <sub>±0.03</sub>       | 0.3748 <sub>±0.01</sub>       | 0.5096 <sub>±0.01</sub>       | 5.3846 <sub>±0.05</sub>       | 3.9907 <sub>±0.04</sub>       | 0.3967 <sub>±0.02</sub>       | 0.5208 <sub>±0.01</sub>       |
| CS-model                     | Ours       | <b>4.9589<sub>±0.03</sub></b> | <b>3.7581<sub>±0.03</sub></b> | <b>0.3796<sub>±0.01</sub></b> | <b>0.5172<sub>±0.01</sub></b> | <b>5.2274<sub>±0.04</sub></b> | <b>3.8788<sub>±0.04</sub></b> | <b>0.4025<sub>±0.02</sub></b> | <b>0.5312<sub>±0.01</sub></b> |
| CPG0016                      |            |                               |                               |                               |                               |                               |                               |                               |                               |
| # Existing reference methods |            |                               |                               |                               |                               |                               |                               |                               |                               |
| TabR                         | ICLR'24    | 2.6335 <sub>±0.05</sub>       | 2.3924 <sub>±0.05</sub>       | 0.0896 <sub>±0.01</sub>       | 0.0865 <sub>±0.01</sub>       | 2.5259 <sub>±0.01</sub>       | 2.2874 <sub>±0.01</sub>       | 0.1330 <sub>±0.01</sub>       | 0.1309 <sub>±0.01</sub>       |
| RealMLP                      | NeurIPS'24 | 2.5152 <sub>±0.04</sub>       | 2.2456 <sub>±0.04</sub>       | 0.3776 <sub>±0.02</sub>       | 0.3579 <sub>±0.02</sub>       | 2.3456 <sub>±0.01</sub>       | 2.0846 <sub>±0.01</sub>       | 0.4658 <sub>±0.01</sub>       | 0.4504 <sub>±0.01</sub>       |
| RF-TD                        | NeurIPS'24 | 2.5885 <sub>±0.05</sub>       | 2.3130 <sub>±0.04</sub>       | 0.0170 <sub>±0.01</sub>       | 0.0224 <sub>±0.01</sub>       | 2.4955 <sub>±0.01</sub>       | 2.2221 <sub>±0.01</sub>       | 0.0758 <sub>±0.01</sub>       | 0.0835 <sub>±0.01</sub>       |
| □ Search-based baselines     |            |                               |                               |                               |                               |                               |                               |                               |                               |
| FLAML                        | MLSys'21   | 2.1126 <sub>±0.03</sub>       | 1.9075 <sub>±0.03</sub>       | 0.5062 <sub>±0.01</sub>       | 0.4974 <sub>±0.01</sub>       | 2.0249 <sub>±0.02</sub>       | 1.8186 <sub>±0.02</sub>       | 0.5614 <sub>±0.02</sub>       | 0.5537 <sub>±0.02</sub>       |
| Random Search                | JMLR'24    | 2.2484 <sub>±0.04</sub>       | 2.0293 <sub>±0.04</sub>       | 0.4385 <sub>±0.02</sub>       | 0.4262 <sub>±0.02</sub>       | 2.1227 <sub>±0.02</sub>       | 1.9084 <sub>±0.02</sub>       | 0.5126 <sub>±0.02</sub>       | 0.5028 <sub>±0.02</sub>       |
| △ Agent-designed methods     |            |                               |                               |                               |                               |                               |                               |                               |                               |
| CellForge                    | arXiv'25   | 2.1038 <sub>±0.03</sub>       | 1.8937 <sub>±0.03</sub>       | 0.4978 <sub>±0.01</sub>       | 0.4896 <sub>±0.01</sub>       | 1.9426 <sub>±0.02</sub>       | 1.7438 <sub>±0.02</sub>       | 0.5867 <sub>±0.03</sub>       | 0.5789 <sub>±0.03</sub>       |
| CS-model                     | Ours       | <b>1.8127<sub>±0.02</sub></b> | <b>1.6328<sub>±0.01</sub></b> | <b>0.5924<sub>±0.01</sub></b> | <b>0.5890<sub>±0.01</sub></b> | <b>1.7303<sub>±0.02</sub></b> | <b>1.5517<sub>±0.02</sub></b> | <b>0.6529<sub>±0.03</sub></b> | <b>0.6518<sub>±0.03</sub></b> |

On both datasets, M4 achieved the highest R-SR, Avg PCC, and Best PCC, and the lowest runtime among the tested open-workflow configurations. Figure 5 records exploratory regressions and validation-retained improvements, linking these aggregate results to the case-study trajectories.

## F.2 RESPONSE-SENSITIVE MORPHOLOGY METRICS

Table 25 describes the observed method differences on the most perturbation-responsive morphology features. We report response-feature RMSE and PCC over the top- $K$  responsive features on BBBC036, BBBC047, and CPG0016 under both SMILES-based and plate-based splits.

## F.3 SINGLE-CELL PERTURBATION PREDICTION

Table 26 reports task-specific workflow results on Norman and Papalex, extending the evaluation beyond morphology and LINCS-style readouts. Norman evaluates transcriptional responses to gene activation;

Papalexι evaluates RNA and protein responses as separate prediction tasks. Each comparison uses its dataset-specific protocol and reports PCC and  $PCC_{DE}$ .

Table 26: **Single-cell perturbation prediction.** We compare task-specific CS-model predictors with CellForge outputs on genetic and multimodal single-cell benchmarks using PCC and  $PCC_{DE}$ . Entries are point estimates under their dataset-specific protocols; the source records do not provide a common replicate unit for a cross-method dispersion term, so no  $\pm$  value is inferred. Best point estimates are **bolded**.

| Dataset                                              | Method          | PCC $\uparrow$ | $PCC_{DE}$ $\uparrow$ |
|------------------------------------------------------|-----------------|----------------|-----------------------|
| <i>CRISPRa Gene-Activation<br/>scRNA-seq Dataset</i> |                 |                |                       |
| Norman                                               | CellForge       | 0.9846         | 0.8109                |
|                                                      | <b>CS-model</b> | <b>0.9871</b>  | <b>0.9060</b>         |
| <i>CRISPR-Cas9 ECCITE-seq<br/>RNA Dataset</i>        |                 |                |                       |
| Papalexι RNA                                         | CellForge       | 0.6935         | 0.6406                |
|                                                      | <b>CS-model</b> | <b>0.8193</b>  | <b>0.8311</b>         |
| <i>CRISPR-Cas9 ECCITE-seq<br/>Protein Dataset</i>    |                 |                |                       |
| Papalexι Protein                                     | CellForge       | 0.7495         | 0.7409                |
|                                                      | <b>CS-model</b> | <b>0.7941</b>  | <b>0.8275</b>         |